CPxplorer

https://github.com/WBS-TW/CPxplorer

V.1.0

A platform-independent toolkit for rapid quantification and harmonization of polychlorinated alkanes

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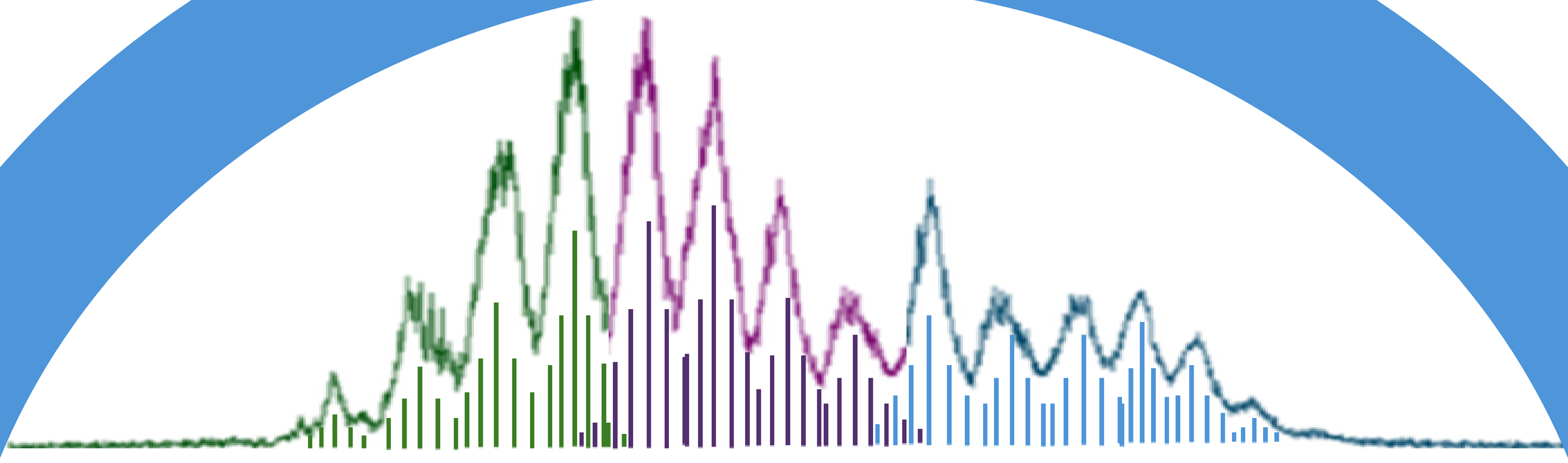
Contributors: Valentina Araya Piqué

This tutorial provides guidance on the workflow to quantify PCAs and related compounds. The interfaces and figures in this tutorial might differ slightly from the latest versions.

Example datasets available here: https://zenodo.org/records/19428486



CPxplorer

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WORKFLOW

1 CREATE A TARGET LIST

CPions

CPions is a graphical user interface that allows exploring mass spectral interferences and creating a target ion list, which excludes the interfering m/z values

2 INTEGRATE THE DATA

Skyline

Skyline is open-source software that allows integrating the mass spectrometric data based on the target list created by CPions and does not necessitate prior data conversion

3 QUANTIFY THE DATA

CPquant

CPquant is a graphical user interface that allows quantifying the integration results from Skyline using the homologue deconvolution method

CPxplorer

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CPions

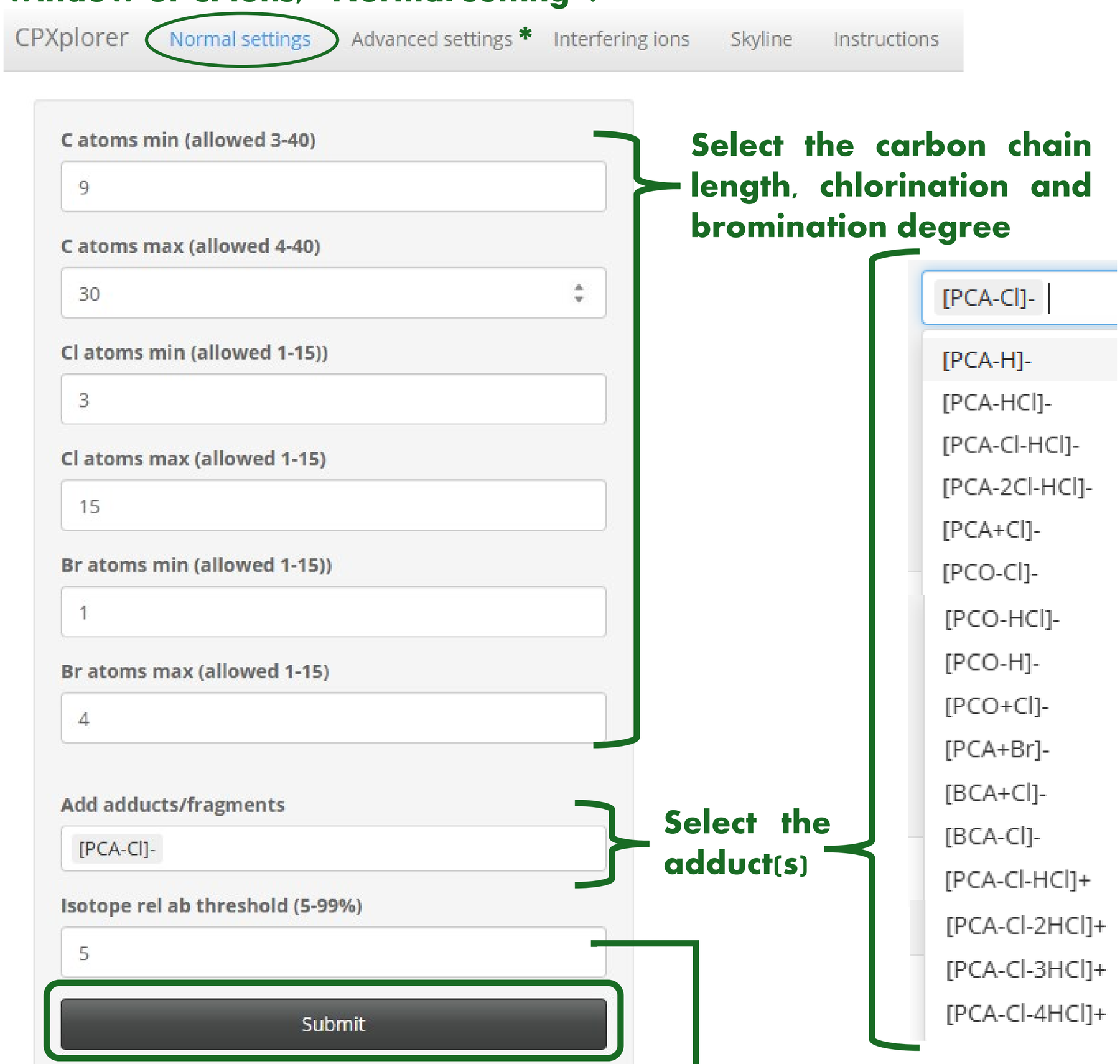
CPxplorer
V.1.0

CREATE A TARGET LIST

CPions is a module part of the CPxplorer package. It consists of an Rstudio based app whose code source can be found at https://github.com/WBS-TW/CPxplorer. It can also be opened directly on the browser at https://wbs-tw.github.io/CPions_Shinylive/, and this method does not require to be downloaded and installed. More instructions can be found in the app.

CPions is a tool for exploring mass spectral interferences of compounds that are present at chlorinated paraffin (CP) mixtures, including polychlorinated alkanes (PCAs), brominated alkanes (BCAs), polychlorinated olefins (PCOs) and PCA- transformation products.

The user can select the compounds to be investigated at the initial window of CPions, "Normal setting":



Select the carbon chain length, chlorination and bromination degree

Select the adduct(s)

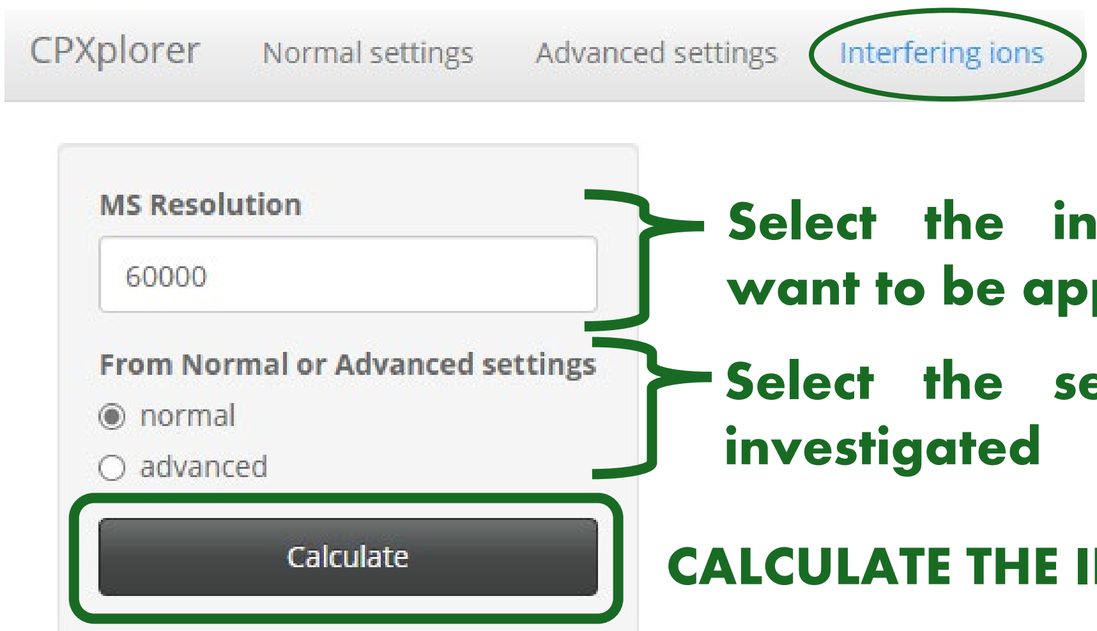
CALCULATE THE INTERFERENCES

Only the m/z values that have a relative abundance above the threshold will be generated

*** BCAs and PCA transformation products can be generated at the "Advanced setting" window**

CREATE A TARGET LIST

The ions that fit the selected parameters will be generated and compiled in a table at the "Normal settings" window. The interferences between the m/z values can be investigated at the "Interfering ions" window:

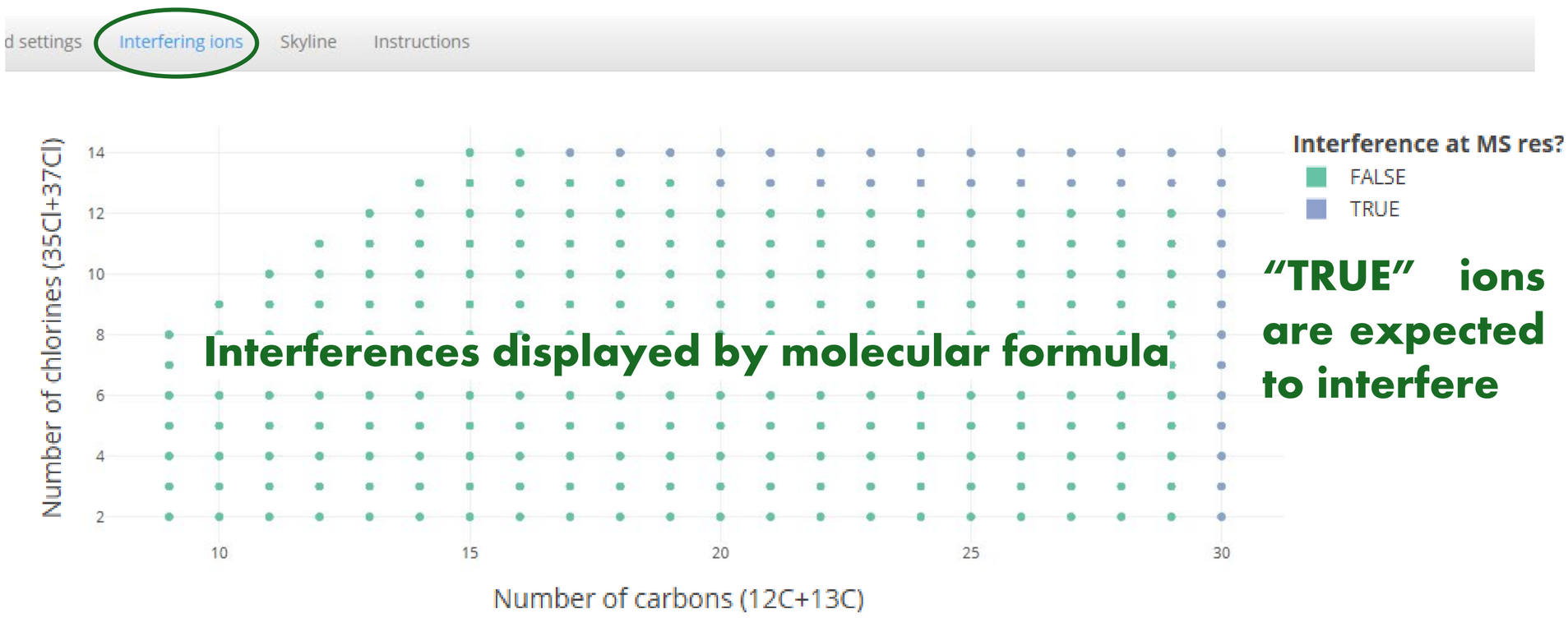


Select the instrumental resolution that want to be applied

Select the settings that want to be investigated

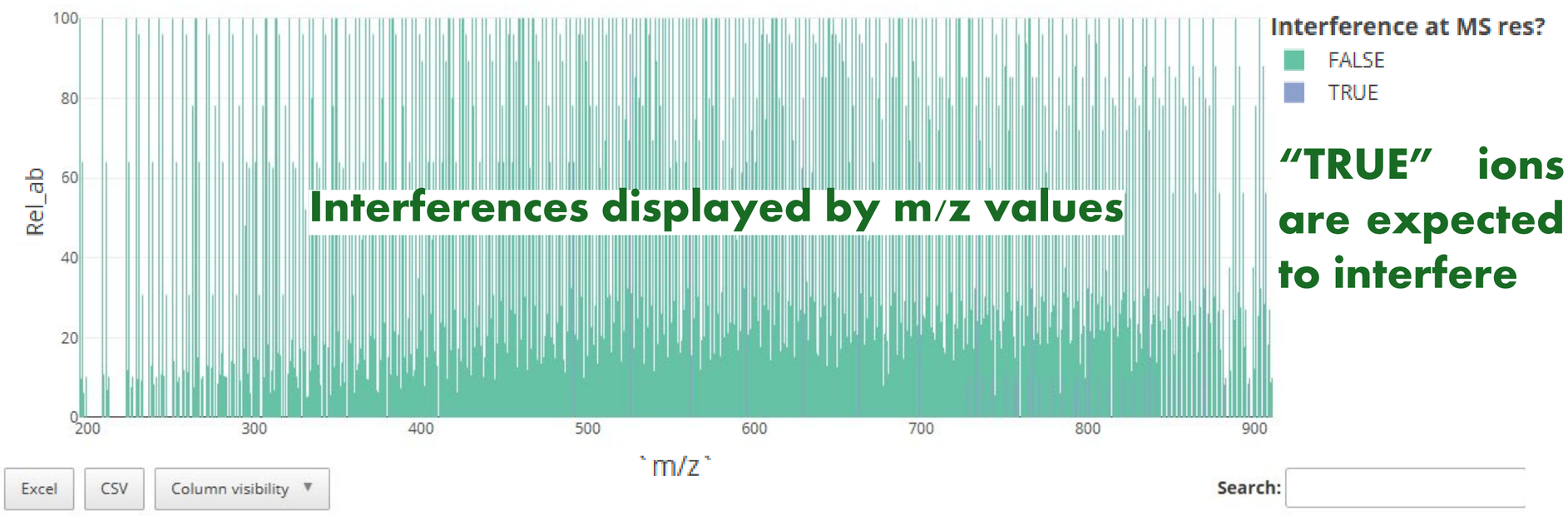
CALCULATE THE INTERFERENCES

The interferences will be displayed in interactive graphs and table:



Interferences displayed by molecular formula

"TRUE" ions are expected to interfere



Interferences displayed by m/z values

"TRUE" ions are expected to interfere

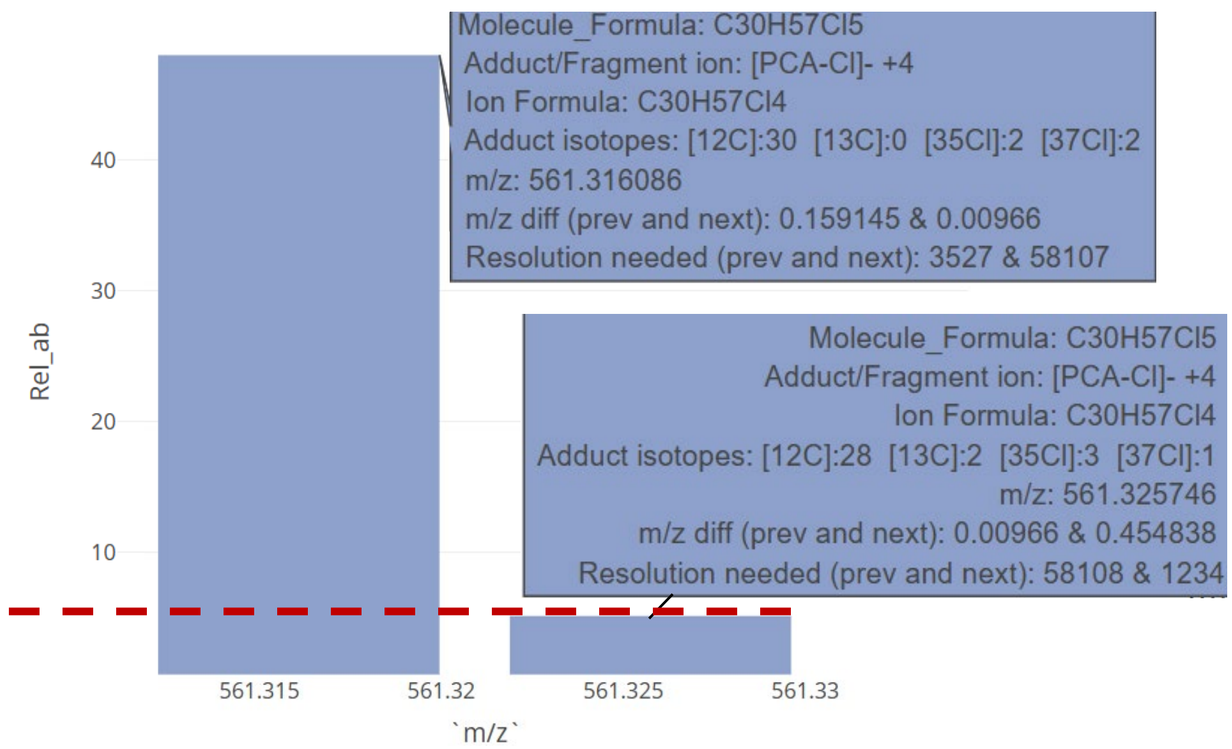
Excel | CSV | Column visibility | Search:

Molecule_Formula	Halo_perc	Charge	Adduct	Adduct_Formula	Isotopologue	Isotope_Formula	m/z	Rel_ab
C9H17Cl3	46	-1	[PCA-Cl]-	C9H17Cl2		[12C]9[13C]0[1H]17[2H]0[35Cl]2[37Cl]0	195.07128	100
C9H17Cl3	46	-1	[PCA-Cl]- +1	C9H17Cl2	+1	[12C]8[13C]1[1H]17[2H]0[35Cl]2[37Cl]0	196.074634	9.7
C9H17Cl3						37Cl]1	197.068329	64
C9H17Cl3	46	-1	[PCA-Cl]- +3	C9H17Cl2	+3	[12C]8[13C]1[1H]17[2H]0[35Cl]1[37Cl]1	198.071684	6.2
C9H17Cl3	46	-1	[PCA-Cl]- +4	C9H17Cl2	+4	[12C]9[13C]0[1H]17[2H]0[35Cl]0[37Cl]2	199.065379	10.2

Compilation of all ions and their interferences

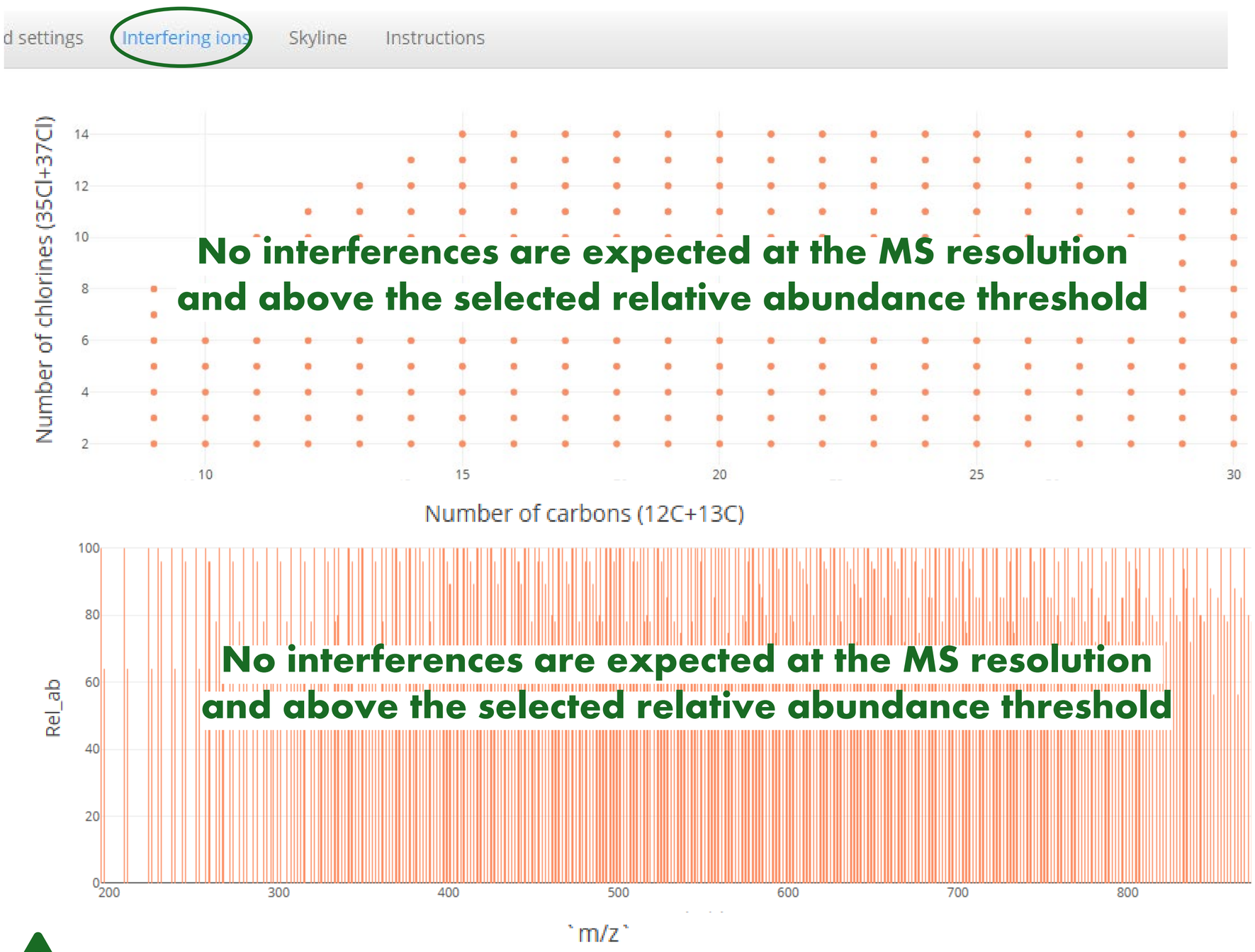
CREATE A TARGET LIST

The graphs can be zoomed in and more information about the interfering ions will be displayed. This allows the user to understand the interferences better and select the target ions. As explained in the following example:



Some of the $[C_{30}H_{57}Cl_4$-Cl]$^-$ +4 isotopes can interfere with each other. The ion with m/z 562.325746 has a relative abundance below 10%. If the user sets the threshold in the previous tab above 10%, it will exclude this ion from the target list.

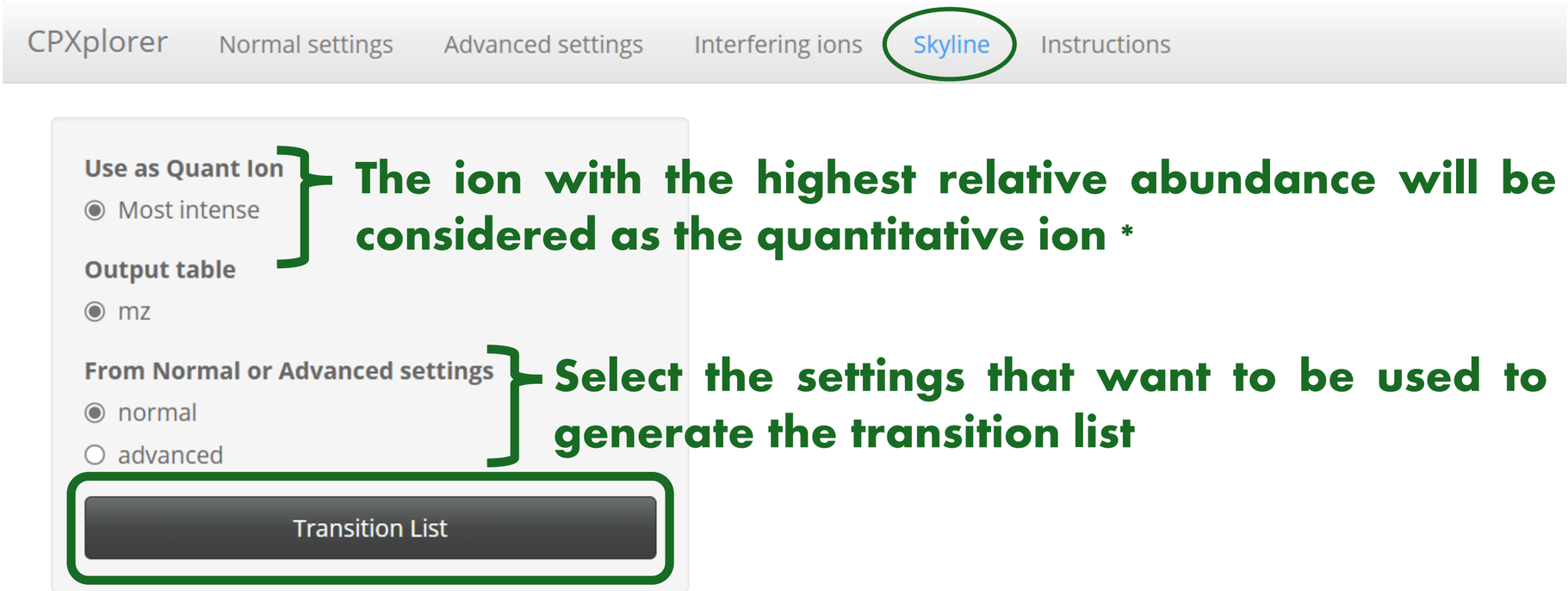
The theoretical relative abundances of some interfering ions are very low or not expected to be present in the samples, making them unlikely to be significant. The user can set a relative abundance threshold higher than that of these ions, ensuring that at least three isotopologues are considered for each homologue group:



Be mindful that interfering ions below the user threshold can still be present in the samples.

CREATE A TARGET LIST

Once the user has selected the target ions, the target list can be created at the "Skyline" window.



GENERATE THE TARGET LIST

The target list can be modified by the user, and it can exported to Excel or CSV files.

Excel | CSV | Column visibility ▼ | Search:

Molecule List Name	Molecule Name	Precursor Charge	Label Type	Precursor m/z	Explicit Retention Time	Explicit Retention Time Window	Note
All	All	All		All	All	All	All
PCA-C11	C11H19Cl5	-1	Qual	296.02209			[PCA-Cl]- -
PCA-C11	C11H19Cl5	-1	Qual	297.015785			[PCA-Cl]- -
PCA-C11	C11H18Cl6	-1	Qual	324.985663			[PCA-Cl]-
PCA-C11	C11H18Cl6	-1	Qual	325.989018			[PCA-Cl]- -
PCA-C11	C11H18Cl6	-1	Quan	326.982713			[PCA-Cl]- -
PCA-C11	C11H18Cl6	-1	Qual	327.986067			[PCA-Cl]- -
PCA-C11	C11H18Cl6	-1	Qual	328.979762			[PCA-Cl]- -
PCA-C11	C11H18Cl6	-1	Qual	329.983117			[PCA-Cl]- -
PCA-C11	C11H18Cl6	-1	Qual	330.976812			[PCA-Cl]- -
PCA-C11	C11H17Cl7	-1	Qual	358.94669			[PCA-Cl]-

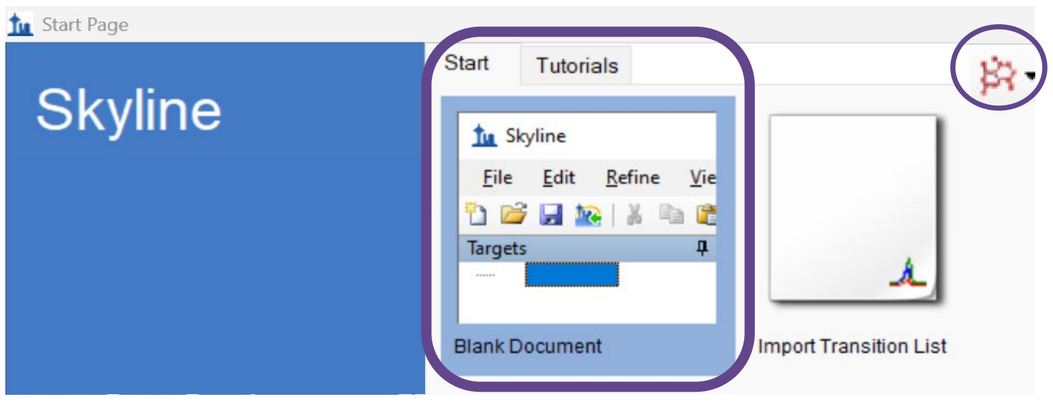
The list including the target compounds is named "Transition list" in Skyline it will be used to extract the ion chromatograms from the raw data in the next step.

*** The user can manually change Quan/Qual in the exported file**

INTEGRATE THE DATA

The next step of the CPxplorer workflow is conducted in Skyline, an open-source Windows client software for treating mass spectrometric data. Skyline can be freely downloaded at the following link: https://skyline.ms/project/home/software/skyline/begin.view. The following sections provide instructions on using Skyline 24.1..

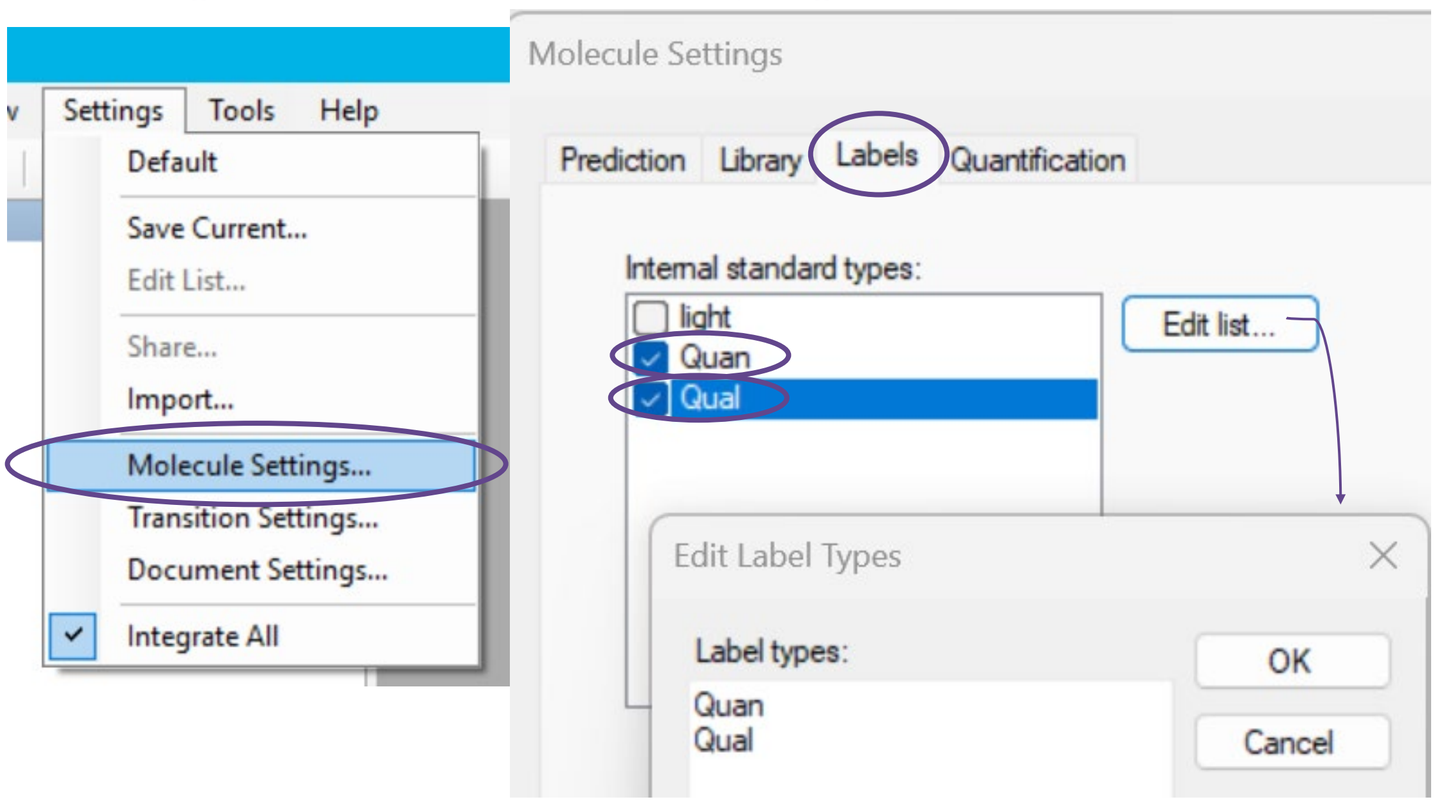
Since Skyline can be used for the analysis of proteomics and small molecules, the first step is to set the software interface to "Molecule interface". Then a Blank Document can be created:



Set the software interface to "Moleclule interface"

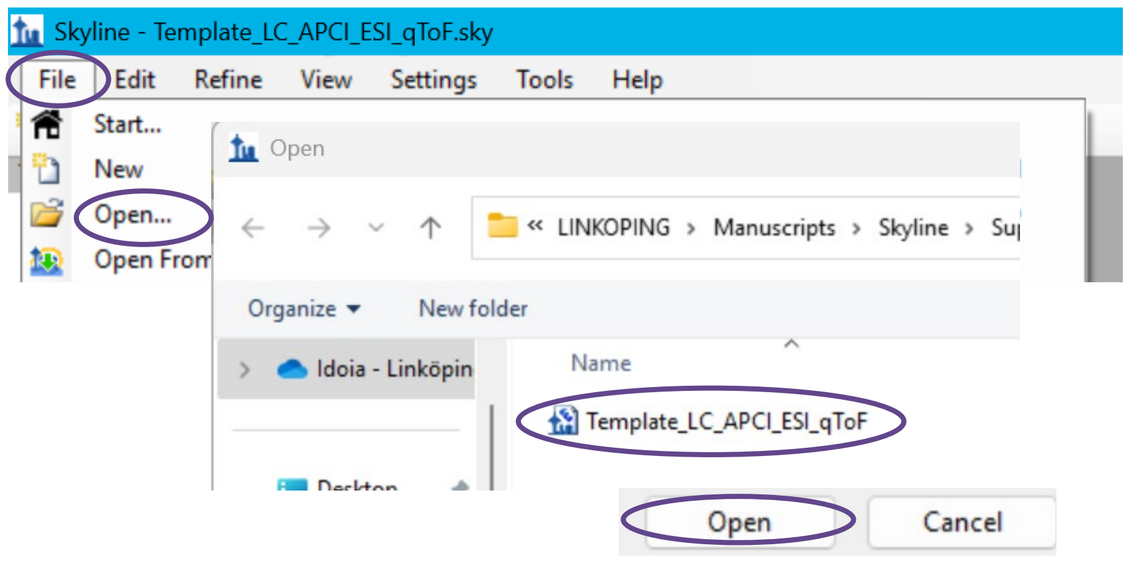
Create a Blank Document

Prior to updating the data into Skyline, the settings for extracting the data must be selected. The "Molecule setting" offers the possibility of labeling the ions and identifying which one is the quantitative one.



Select the labels "Quan" and "Qual", if they do not appear on the display, "Edit list" and manually add these two labels

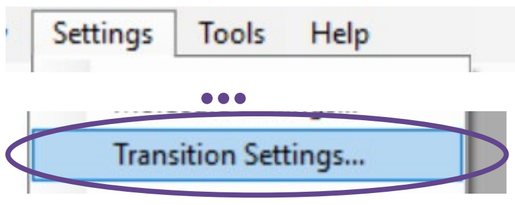
Instrumental settings can be user-defined (see page 6); however presets (.sky format) for data from different chromatographic and mass spectrometric instruments can be found in the /inst folder at: https://github.com/WBS-TW/CPxplorer.

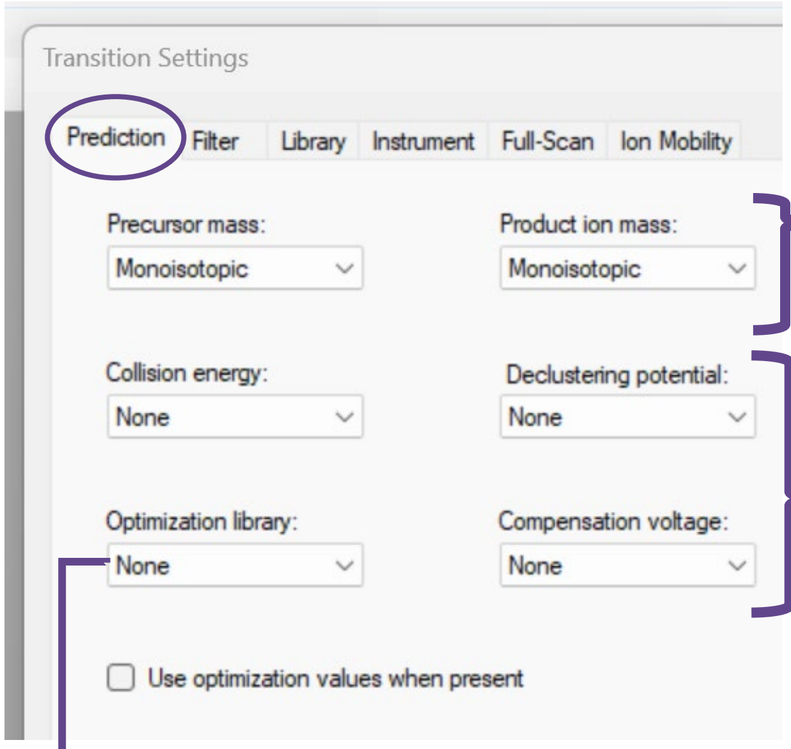


The method can be opened in Skyline. The settings will be then set, and the .raw data can be directly uploaded into Skyline. The multipanel display will also automatically appear when the data is loaded.

INTEGRATE THE DATA

The user can modify the instrumental parameters at the "Setting" -> "Transition setting" window. Below are some instructions.

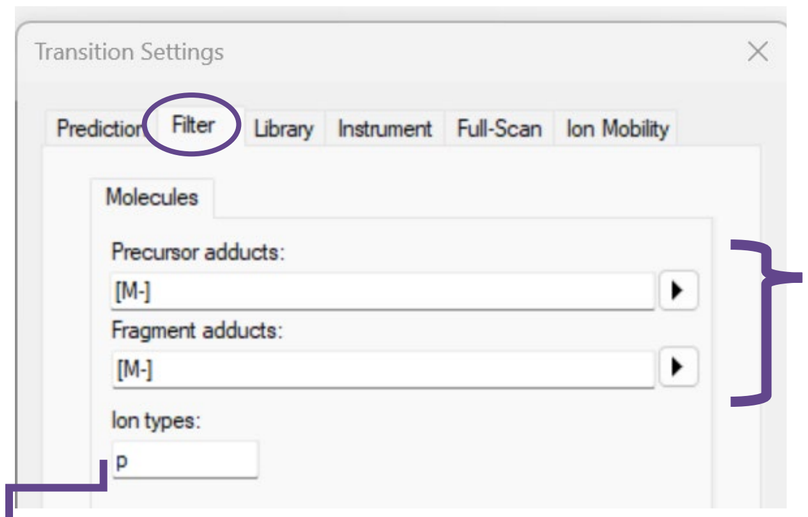




Not available library

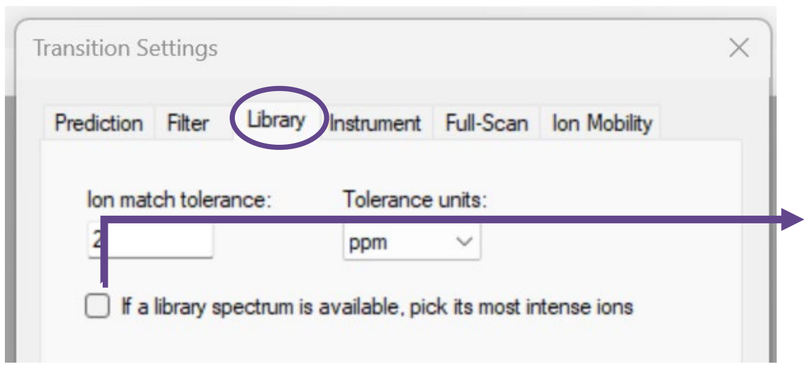
The transition list generated by CPions contains several ions corresponding to different isotopes of each molecule. Therefore, the "precursor mass" and "product ion mass" can be both set as "monoisotopic"

CP mixtures are analyzed only using MS_1 data. Consequently, the "Collision energy", "Declustering potential", and "Compensation voltage" can be set as "None".

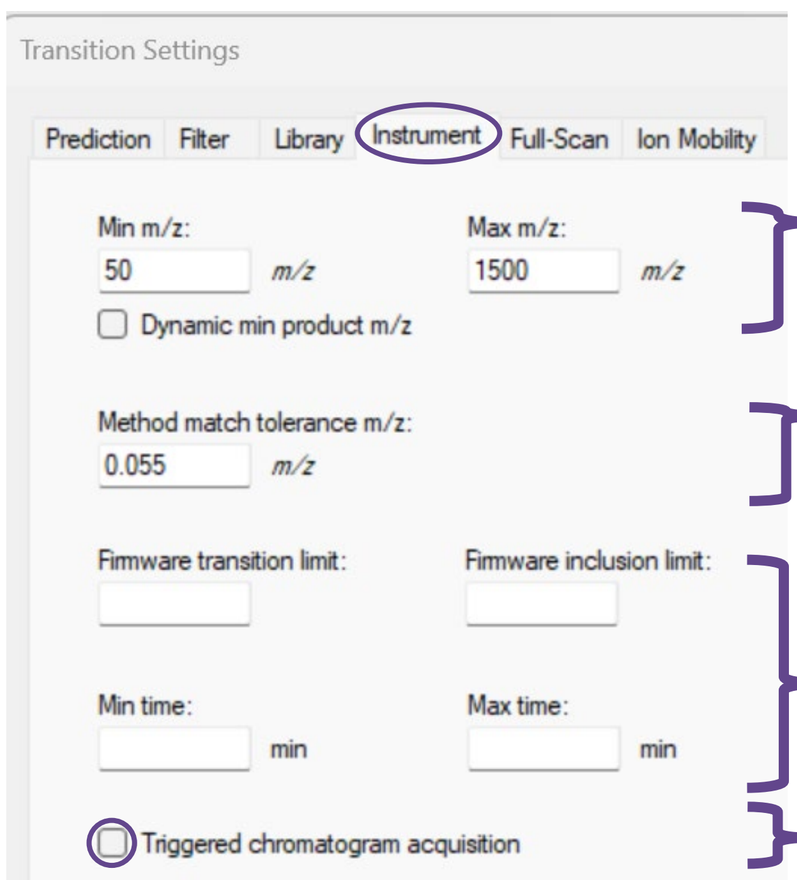


The adducts were already selected in CPions, and they are included in the transition list. So, the "Precursor" and "Fragment" adducts can be set as [M-] and [M+] when applying a negative and positive ionization technique, respectively.

"Ion types" can be set as "p" (precursor) when only MS_1 data is used



Since a spectral library is not used keep the library box unselected



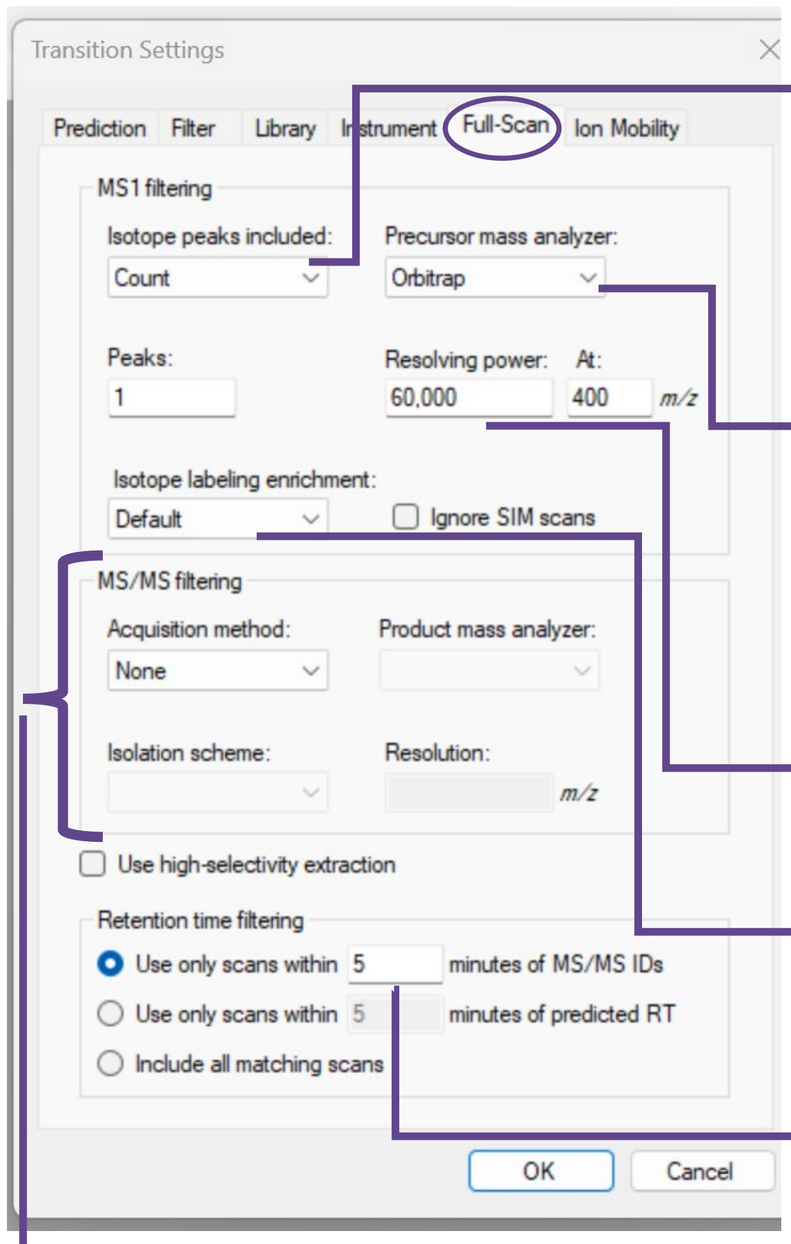
m/z range from the scan used for acquiring the data

Mass error acceptancy when extracting the chromatograms. 0.005 and 0.010 are recommended for Orbitrap and qToF data, respectively

The firmware is not needed when using MS_1 data solely. It is a multiplexing strategy for data-independent acquisition (DIA)-based mass spectrometry data,

Keep this box deselected

INTEGRATE THE DATA



The "Count" option can be selected, to get a specific value for each "Peaks"

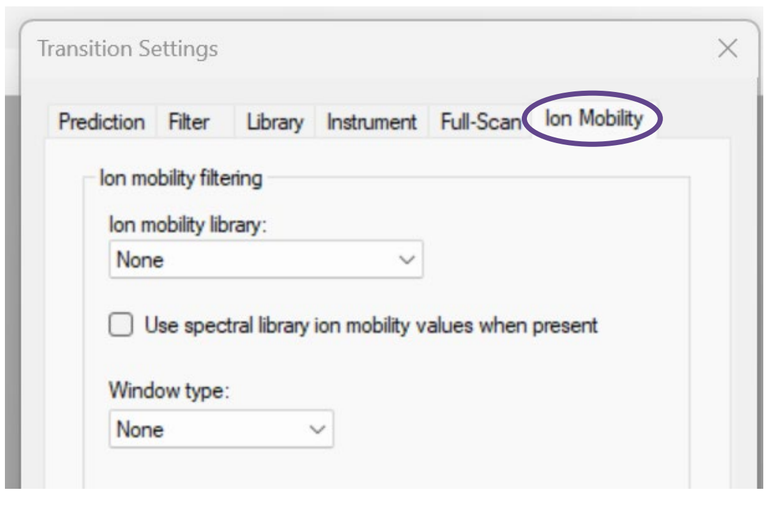
Selecting the instrument used will prompt Skyline to extract chromatograms from profile data, summing all profile intensities across an m/z range within the specified mass resolution

The instrumental resolution used for acquiring the data

The isotope labeling can be set as default because it was already indicated in the "Molecule setting"

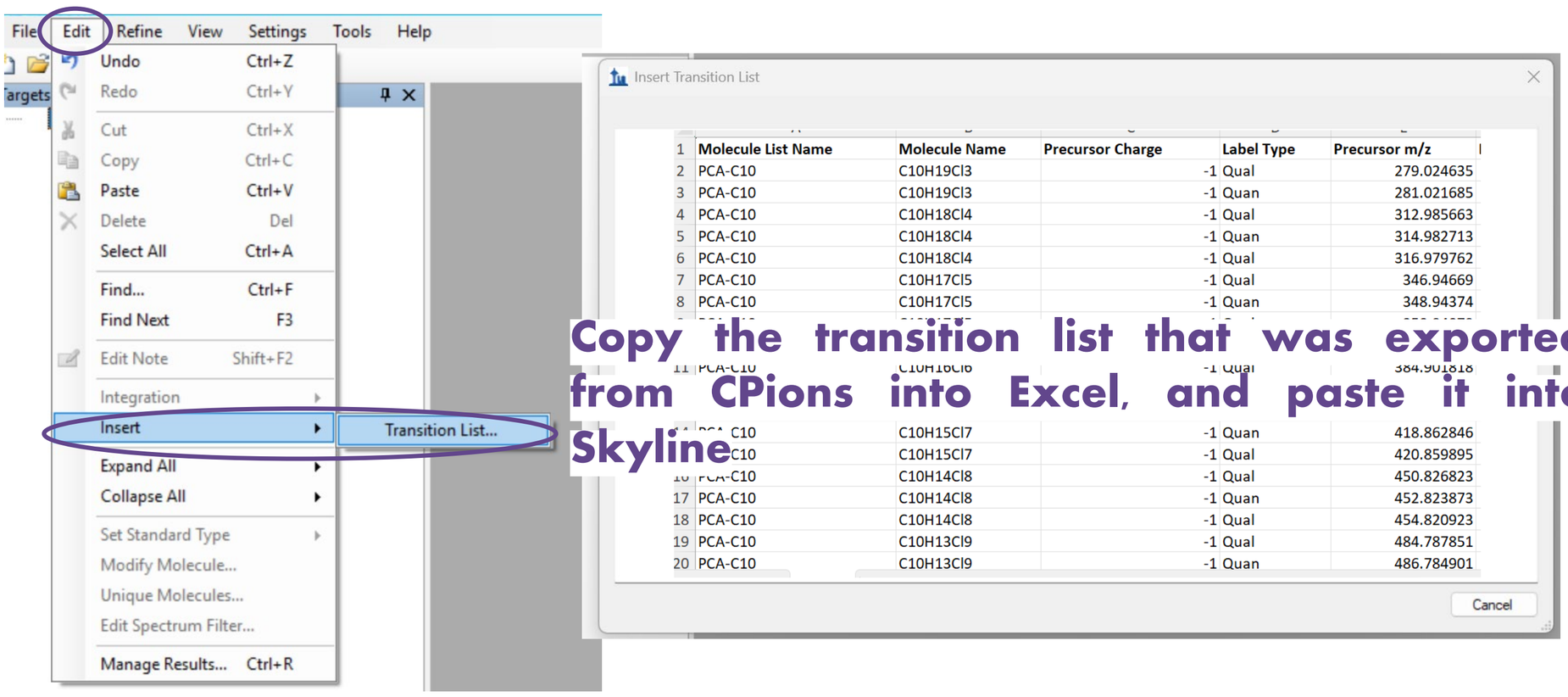
Set a broad window, ex.: 5 min

When only MS_1 data is used the "MS/MS filtering" can be set as "none"



If the data also includes an ion mobility dimension, the library and window type can be indicated

Once the settings are introduced, the transition list can be imported

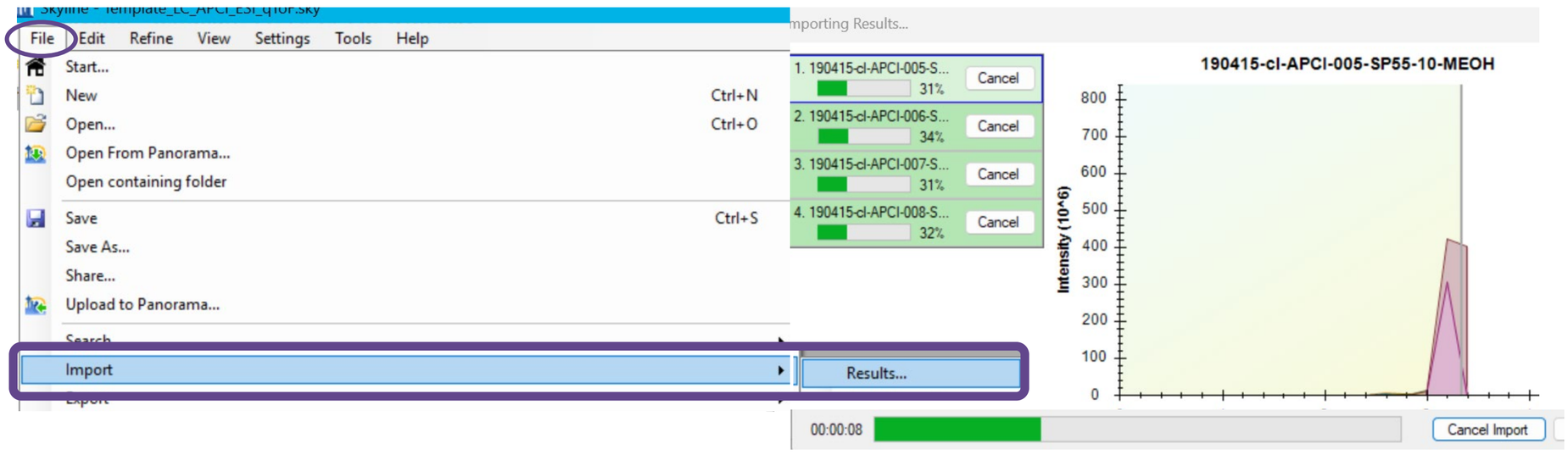


	Molecule List Name	Molecule Name	Precursor Charge	Label Type	Precursor m/z
2	PCA-C10	C10H19Cl3	-1	Qual	279.024635
3	PCA-C10	C10H19Cl3	-1	Quan	281.021685
4	PCA-C10	C10H18Cl4	-1	Qual	312.985663
5	PCA-C10	C10H18Cl4	-1	Quan	314.982713
6	PCA-C10	C10H18Cl4	-1	Qual	316.979762
7	PCA-C10	C10H17Cl5	-1	Qual	346.94669
8	PCA-C10	C10H17Cl5	-1	Quan	348.94374
11	PCA-C10	C10H16Cl6	-1	Qual	384.901818
14	PCA-C10	C10H15Cl7	-1	Quan	418.862846
15	PCA-C10	C10H15Cl7	-1	Qual	420.859895
16	PCA-C10	C10H14Cl8	-1	Qual	450.826823
17	PCA-C10	C10H14Cl8	-1	Quan	452.823873
18	PCA-C10	C10H14Cl8	-1	Qual	454.820923
19	PCA-C10	C10H13Cl9	-1	Qual	484.787851
20	PCA-C10	C10H13Cl9	-1	Quan	486.784901

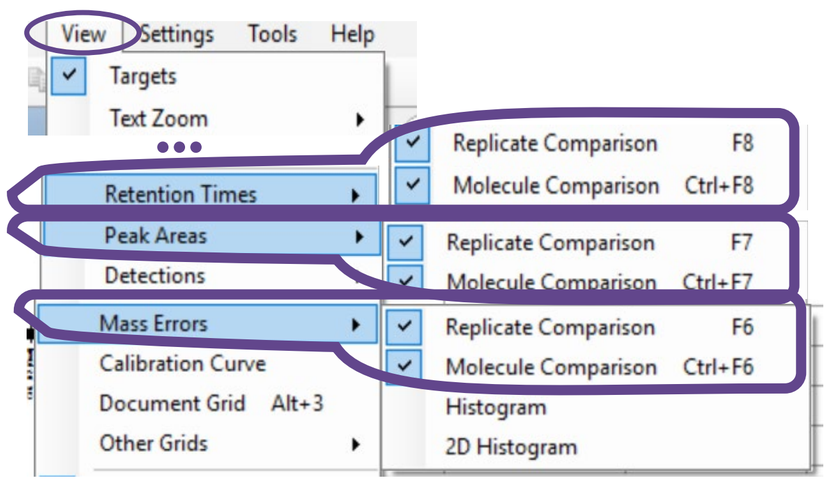
Copy the transition list that was exported from CPions into Excel, and paste it into Skyline

INTEGRATE THE DATA

Once the settings are set, the method can be saved and the raw data imported. Skyline supports major MS vendor raw files as well as .mzml format. without prior data conversion.

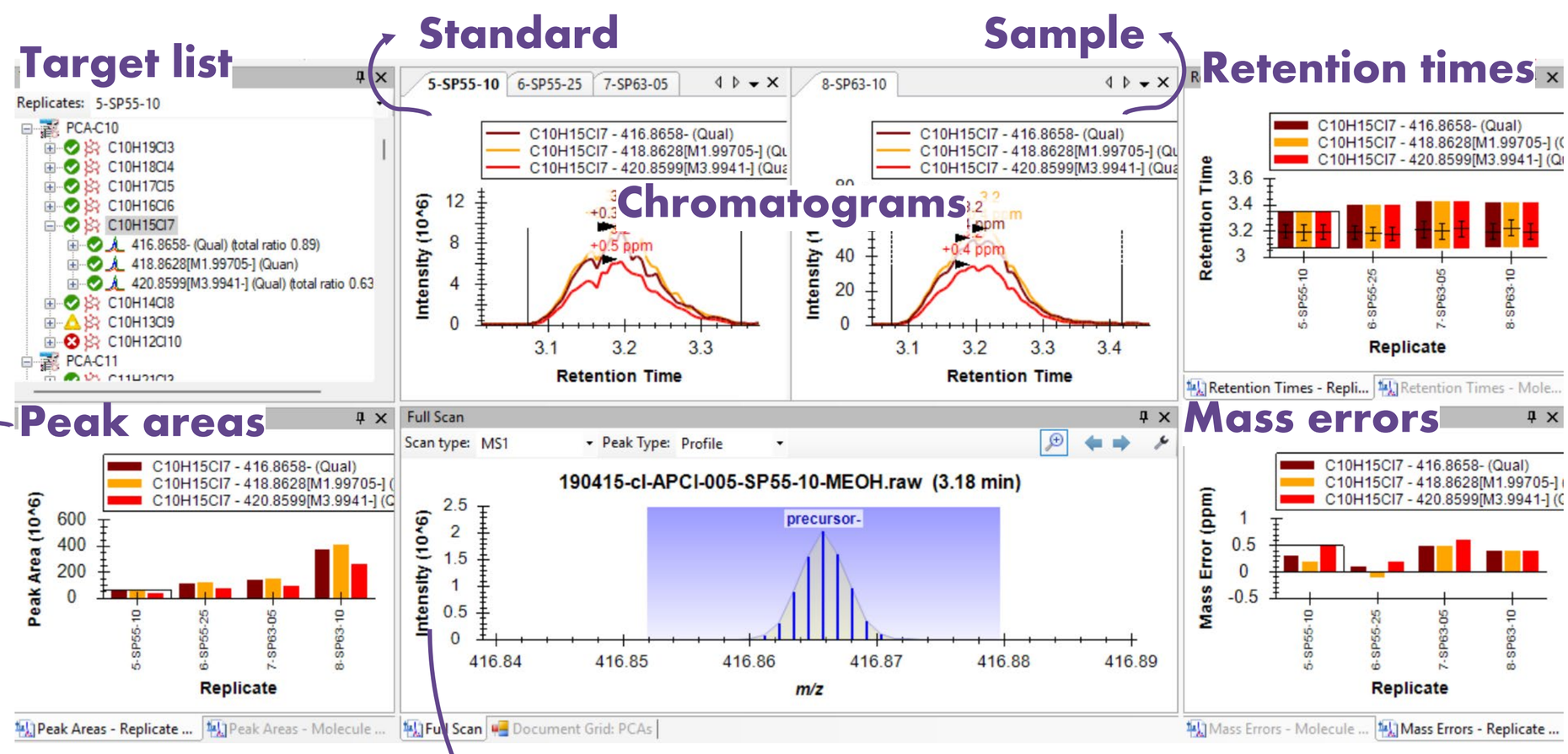


For the peak integration, the CPxplorer workflow advises verifying several parameters, including retention time, isotopic distribution, mass error, and m/z profile data distribution. Therefore, several panels are recommended to display in Skyline



The multiparameter display can be selected at the window "View". The "Replicate Comparison" option displays the same Molecule for the different samples; while the "Molecule Comparison" shows all the molecules in all the samples

The display should look like the following:



It shows the isotopic distribution

The profile data will be displayed when clicking on the chromatogram of one of the ions

Note: when opening one of the methods available in the CPxplorer GitHub repository, the display appears automatically.

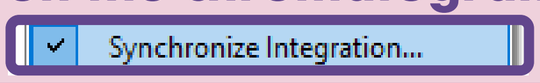
INTEGRATE THE DATA

Once the Skyline display is ready, the peaks can be integrated: The chromatograms of the different targets can be displayed by clicking on them in the transition list, and the peaks are integrated by selecting the start and end point of the chromatogram in the retention time axis. The following criteria is advised for selecting the data:

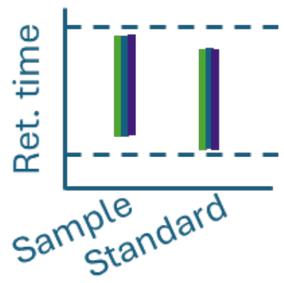
Example:

$C_{18}H_{31}Cl_7$: target m/z
- 528.9910 (qual)
- 530.9880 (quan)
- 532.9851 (qual)

Synchronize the integration of several files by right-clicking on the chromatogram:

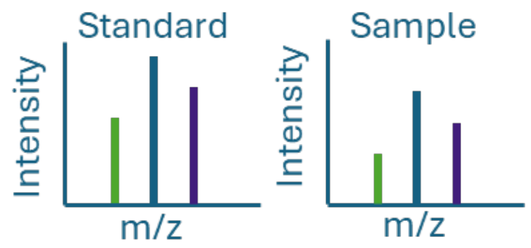


- Retention time:



Same retention time range for the standard and the sample

- Peak area:



Same isotopic distribution for the standard and the sample

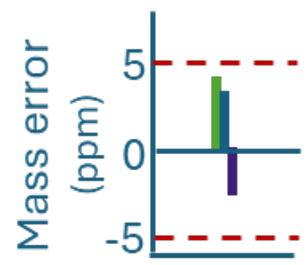
- MS profile data:

MS profile data showing Gaussian distribution

- Chromatograms:

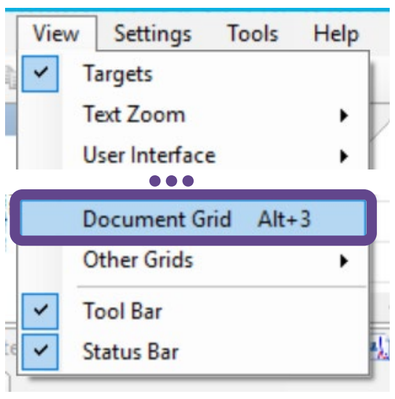
Same peak shape for all isotopomers, the "quan" and "qual" ions

- Mass errors:



For Orbitrap data less than ±5 ppm, and for qToF ±15 ppm

The integrated data is compiled in the "Document Grid", this is a report that can be edited by the user while treating the data simultaneously. It can be displayed in the "View" window:



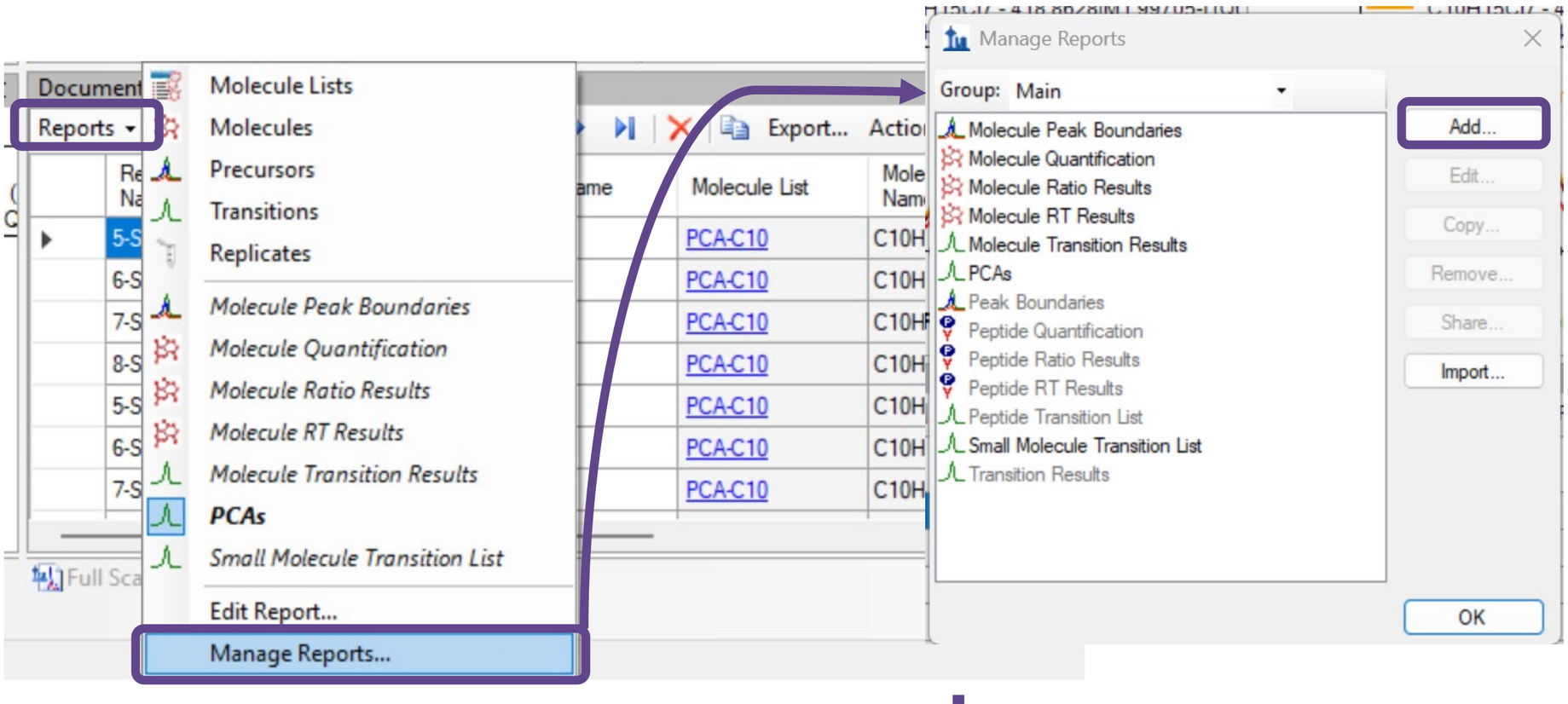
Document Grid

Reports | 34 of 8481 | Export... | Actions | Find:

Replicate Name	Sample Type	Batch Name	Molecule List	Molecule	Area	Analyte Concentration	Mass Error PPM	Isotope Label Type
51_CAL1	Standard	C10-13_A	PCA-C10	C10H19Cl3	0	0.3	#N/A	Quan
51_CAL2	Standard	C10-13_A	PCA-C10	C10H19Cl3	0	1	#N/A	Quan
51_CAL3	Standard	C10-13_A	PCA-C10	C10H19Cl3	0	2	#N/A	Quan
51_CAL4	Standard	C10-13_A	PCA-C10	C10H19Cl3	0	4	#N/A	Quan
51_CAL5	Standard	C10-13_A	PCA-C10	C10H19Cl3	0	16	#N/A	Quan
55_CAL1	Standard	C10-13_B	PCA-C10	C10H19Cl3	0	0.3	#N/A	Quan
55_CAL2	Standard	C10-13_B	PCA-C10	C10H19Cl3	0	1	#N/A	Quan
55_CAL3	Standard	C10-13_B	PCA-C10	C10H19Cl3	0	2	#N/A	Quan
55_CAL4	Standard	C10-13_B	PCA-C10	C10H19Cl3	0	4	#N/A	Quan
55_CAL5	Standard	C10-13_B	PCA-C10	C10H19Cl3	0	16	#N/A	Quan
63_CAL1	Standard	C10-13_C	PCA-C10	C10H19Cl3	0	0.3	#N/A	Quan
63_CAL2	Standard	C10-13_C	PCA-C10	C10H19Cl3	0	1	#N/A	Quan
63_CAL3	Standard	C10-13_C	PCA-C10	C10H19Cl3	0	2	#N/A	Quan
63_CAL4	Standard	C10-13_C	PCA-C10	C10H19Cl3	0	4	#N/A	Quan
63_CAL5	Standard	C10-13_C	PCA-C10	C10H19Cl3	0	16	#N/A	Quan
42_CAL1	Standard	C14-17_D	PCA-C10	C10H19Cl3	0	0.3	#N/A	Quan

INTEGRATE THE DATA

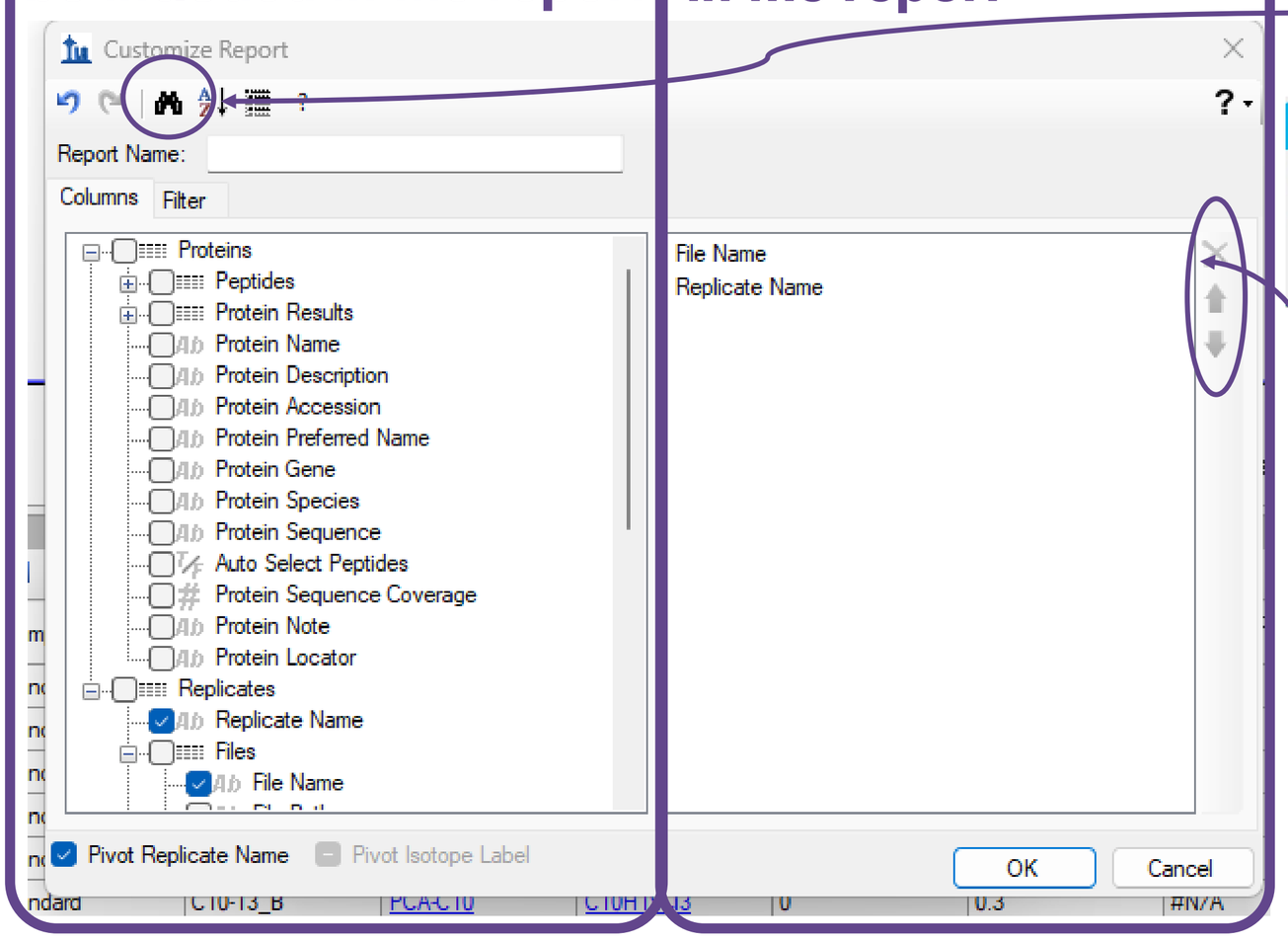
Once the Document Grid is open, a new report can be created in the "Reports" display, by "Manage Reports" and then "Add". Then a window with all the parameters to be included in the report will display and the desired ones can be included in the report.



Select the parameters to be included in the report

Parameters included in the report

Searcher for finding a specific parameter



Organize the parameters from the report using the arrows, and remove the not desired ones by using the cross

Once the new report is created, the document Grid saves automatically. The user can edit, and include data in the document Grid while treating the data, and once all the data analysis is finished a report can be exported. The report will be the same as the document Grid. This report can be exported from Skyline and integrated then into CPquant for quantification without further steps in between.

INTEGRATE THE DATA

In order to be able to integrate the report of Skyline into CPquant, the next step of CPxplorer, some parameters need to be included in the document Grid (Skyline report). Since different versions of Skyline will name these parameters slightly different, the user should make sure the heading are displayed like the demo skyline report found in https://zenodo.org/records/19428486.

The parameters needed are:

-Replicate Name: sample name

-Sample Type: the following characters can be used, Unknown (which is the sample to be quantified), Blank (field blanks and procedural blanks are not distinguished), Standard (standard used for quantification), Quality Control (standard/sample used to determine the recovery).

-Batch Name: for standards only (leave blank for Unknown). This will determine which standards that belongs to a calibration series as well as which carbon chain groups to quantify with the standard. The naming of the Batch Name should be: CarbonGroups_StandardName. An underscore is a separator for the carbon chain group and standard name. Example A: C_{10}-C_{13}_StandardA. This standard will then be used to quantify carbon chains C_{10}, C_{12}, C_{13} (the hyphen specify the range of carbon chains). This belongs to the StandardA which can be at different Analyte Concentration for the same calibration series. Example B: C_{14}_52%Cl. This standard will only quantify C_{14} carbon chains. It specifies 52% chlorine content (although this information is not needed for quantification).

-Molecule List: compounds used internal standards are denoted IS, recovery standards as RS (also called volumetric standard).

-Molecule: PCA homologue group.

-Area: integrated area from Skyline.

-Analyte Concentration: for standards only. This is the standard concentrations/amounts. This column could be in concentration or weight/amount unit depending on the user input. It will affect the final quantification unit.

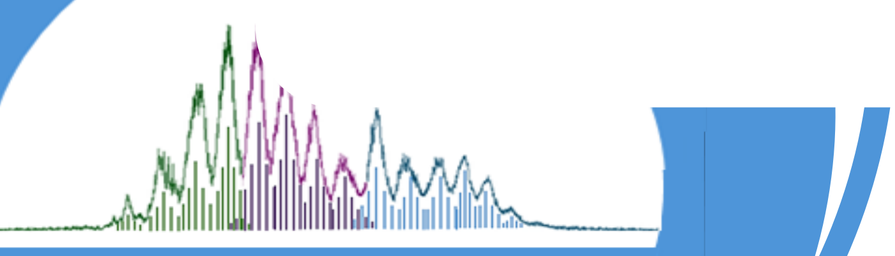
-Mass Error PPM: might be exported from Skyline but currently not used by CPquant.

-Isotope Label Type: Quan or Qual.

-Chromatogram Precursor M/Z: the m/z values of the ion (not used by CPquant).

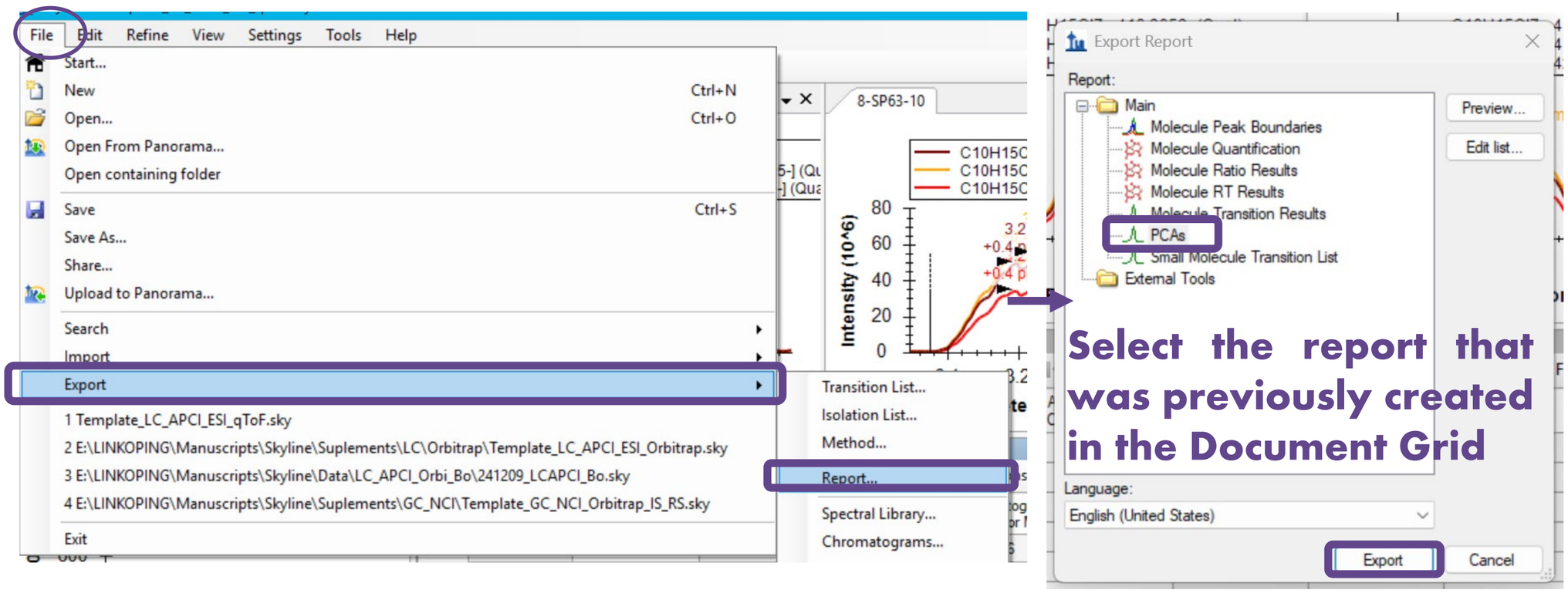
-Sample Dilution Factor: indicate the dilution (>1) or concentration factor (<1). Default from Skyline is 1 (no dilution).

-Transition Note: internal information transferred from CPions used for calculating the correct isotope ratio.

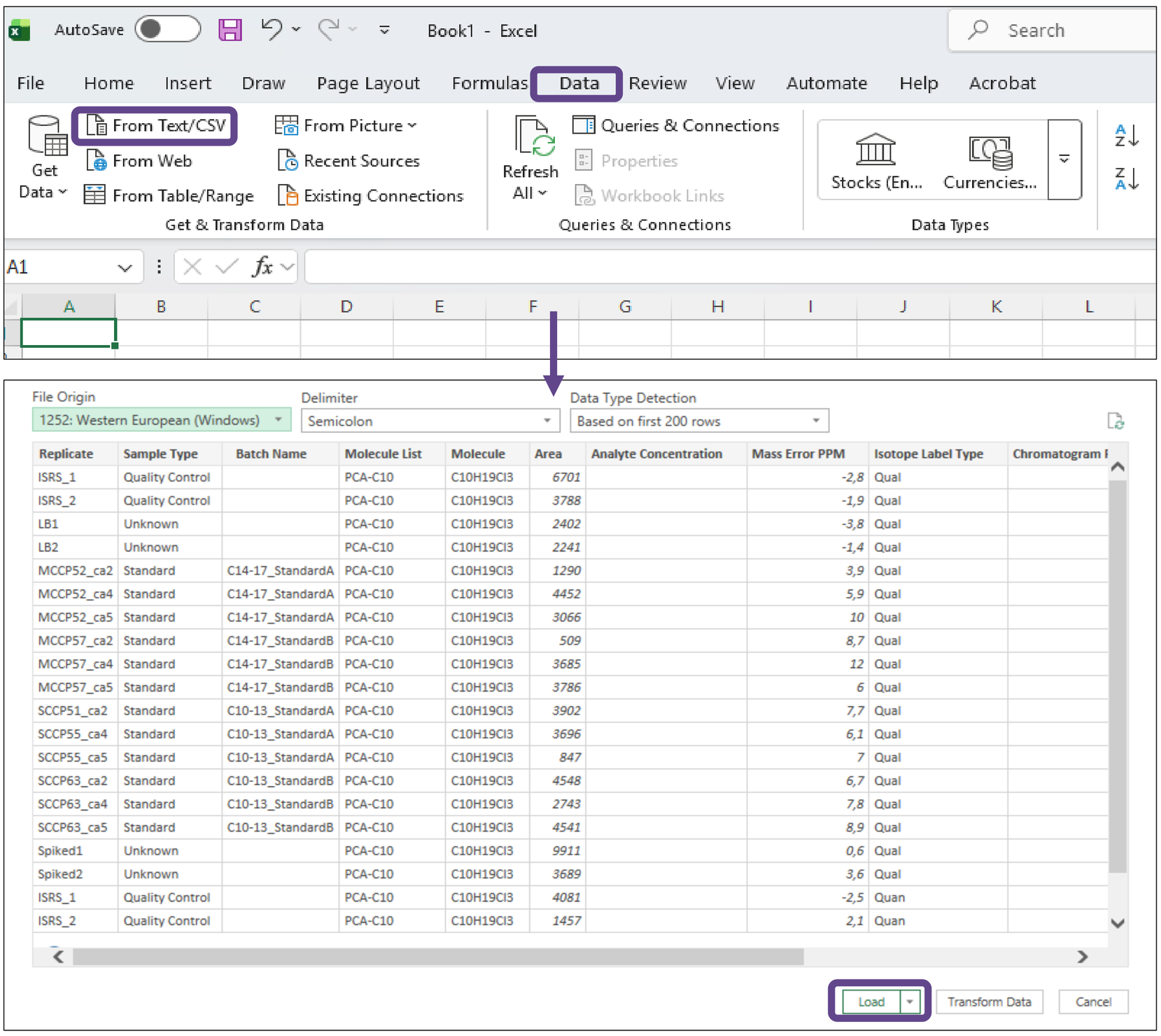


INTEGRATE THE DATA

Once the report is finished, either with these parameters for CPquant or with the user's choice, it can be Exported to CSV or TSV.



If the user intends to use CPquant, it should be converted to .xlsx. This can be done in Excel:



Replicate	Sample Type	Batch Name	Molecule List	Molecule	Area	Analyte Concentration	Mass Error PPM	Isotope Label Type	Chromatogram I
ISRS_1	Quality Control		PCA-C10	C10H19Cl3	6701		-2,8	Qual	
ISRS_2	Quality Control		PCA-C10	C10H19Cl3	3788		-1,9	Qual	
LB1	Unknown		PCA-C10	C10H19Cl3	2402		-3,8	Qual	
LB2	Unknown		PCA-C10	C10H19Cl3	2241		-1,4	Qual	
MCCP52_ca2	Standard	C14-17_StandardA	PCA-C10	C10H19Cl3	1290		3,9	Qual	
MCCP52_ca4	Standard	C14-17_StandardA	PCA-C10	C10H19Cl3	4452		5,9	Qual	
MCCP52_ca5	Standard	C14-17_StandardA	PCA-C10	C10H19Cl3	3066		10	Qual	
MCCP57_ca2	Standard	C14-17_StandardB	PCA-C10	C10H19Cl3	509		8,7	Qual	
MCCP57_ca4	Standard	C14-17_StandardB	PCA-C10	C10H19Cl3	3685		12	Qual	
MCCP57_ca5	Standard	C14-17_StandardB	PCA-C10	C10H19Cl3	3786		6	Qual	
SCCP51_ca2	Standard	C10-13_StandardA	PCA-C10	C10H19Cl3	3902		7,7	Qual	
SCCP55_ca4	Standard	C10-13_StandardA	PCA-C10	C10H19Cl3	3696		6,1	Qual	
SCCP55_ca5	Standard	C10-13_StandardA	PCA-C10	C10H19Cl3	847		7	Qual	
SCCP63_ca2	Standard	C10-13_StandardB	PCA-C10	C10H19Cl3	4548		6,7	Qual	
SCCP63_ca4	Standard	C10-13_StandardB	PCA-C10	C10H19Cl3	2743		7,8	Qual	
SCCP63_ca5	Standard	C10-13_StandardB	PCA-C10	C10H19Cl3	4541		8,9	Qual	
Spiked1	Unknown		PCA-C10	C10H19Cl3	9911		0,6	Qual	
Spiked2	Unknown		PCA-C10	C10H19Cl3	3689		3,6	Qual	
ISRS_1	Quality Control		PCA-C10	C10H19Cl3	4081		-2,5	Quan	
ISRS_2	Quality Control		PCA-C10	C10H19Cl3	1457		2,1	Quan	

INTEGRATE THE DATA

The following REPORT is an example of how the exported report from Skyline should look like, if the user wants to use CPquant:

	A	B	C	D	E	F	G	H	I	J	K	L
1	Replicate	Sample Type	Batch Name	Molecule List	Molecule	Area	Analyte Concentration	Mass Error PPM	Isotope Label Type	Chromatogram Precursor M/Z	Sample Dilution Factor	Transition Note
2	251003_Chlorowax50_ca1	Standard	C10-30_A	PCA-C10	C10H19Cl3	#N/A	50	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
3	251003_Chlorowax50_ca2	Standard	C10-30_A	PCA-C10	C10H19Cl3	#N/A	500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
4	251003_Chlorowax50_ca3	Standard	C10-30_A	PCA-C10	C10H19Cl3	#N/A	5000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
5	251003_Chlorowax50_ca4	Standard	C10-30_A	PCA-C10	C10H19Cl3	#N/A	15000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
6	251003_Chlorowax50_ca5	Standard	C10-30_A	PCA-C10	C10H19Cl3	#N/A	30000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
7	251003_Chlorowax55_ca1	Standard	C10-30_B	PCA-C10	C10H19Cl3	#N/A	50	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
8	251003_Chlorowax55_ca2	Standard	C10-30_B	PCA-C10	C10H19Cl3	#N/A	500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
9	251003_Chlorowax55_ca3	Standard	C10-30_B	PCA-C10	C10H19Cl3	#N/A	5000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
10	251003_Chlorowax55_ca4	Standard	C10-30_B	PCA-C10	C10H19Cl3	#N/A	15000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
11	251003_Chlorowax55_ca5	Standard	C10-30_B	PCA-C10	C10H19Cl3	#N/A	30000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
12	251003_LCCP36_ca1	Standard	C18-C30_A	PCA-C10	C10H19Cl3	#N/A	5	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
13	251003_LCCP36_ca2	Standard	C18-C30_A	PCA-C10	C10H19Cl3	#N/A	50	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
14	251003_LCCP36_ca3	Standard	C18-C30_A	PCA-C10	C10H19Cl3	#N/A	500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
15	251003_LCCP36_ca3b	Standard	C18-C30_A	PCA-C10	C10H19Cl3	#N/A	500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
16	251003_LCCP36_ca4	Standard	C18-C30_A	PCA-C10	C10H19Cl3	#N/A	1500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
17	251003_LCCP36_ca5	Standard	C18-C30_A	PCA-C10	C10H19Cl3	#N/A	3000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
18	251003_LCCP49_ca1	Standard	C18-C30_B	PCA-C10	C10H19Cl3	#N/A	5	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
19	251003_LCCP49_ca2	Standard	C18-C30_B	PCA-C10	C10H19Cl3	#N/A	50	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
20	251003_LCCP49_ca3	Standard	C18-C30_B	PCA-C10	C10H19Cl3	#N/A	500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
21	251003_LCCP49_ca4	Standard	C18-C30_B	PCA-C10	C10H19Cl3	#N/A	1500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
22	251003_LCCP49_ca5	Standard	C18-C30_B	PCA-C10	C10H19Cl3	#N/A	3000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
23	251003_LCCP66_ca1	Standard	C18-C30_C	PCA-C10	C10H19Cl3	#N/A	5	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
24	251003_LCCP66_ca2	Standard	C18-C30_C	PCA-C10	C10H19Cl3	#N/A	50	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
25	251003_LCCP66_ca3	Standard	C18-C30_C	PCA-C10	C10H19Cl3	#N/A	500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
26	251003_LCCP66_ca4	Standard	C18-C30_C	PCA-C10	C10H19Cl3	#N/A	1500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
27	251003_LCCP66_ca5	Standard	C18-C30_C	PCA-C10	C10H19Cl3	#N/A	3000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
28	251003_MCCP42_ca1	Standard	C14-C17_A	PCA-C10	C10H19Cl3	#N/A	5	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
29	251003_MCCP42_ca2	Standard	C14-C17_A	PCA-C10	C10H19Cl3	#N/A	50	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
30	251003_MCCP42_ca3	Standard	C14-C17_A	PCA-C10	C10H19Cl3	#N/A	500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
31	251003_MCCP42_ca4	Standard	C14-C17_A	PCA-C10	C10H19Cl3	#N/A	1500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
32	251003_MCCP42_ca5	Standard	C14-C17_A	PCA-C10	C10H19Cl3	#N/A	3000	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
33	251003_MCCP52_ca1	Standard	C14-C17_B	PCA-C10	C10H19Cl3	#N/A	5	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
34	251003_MCCP52_ca2	Standard	C14-C17_B	PCA-C10	C10H19Cl3	#N/A	50	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}
35	251003_MCCP52_ca3	Standard	C14-C17_B	PCA-C10	C10H19Cl3	#N/A	500	#N/A	Qual	279.0246	1	{[PCA+Cl]-}{78.1}

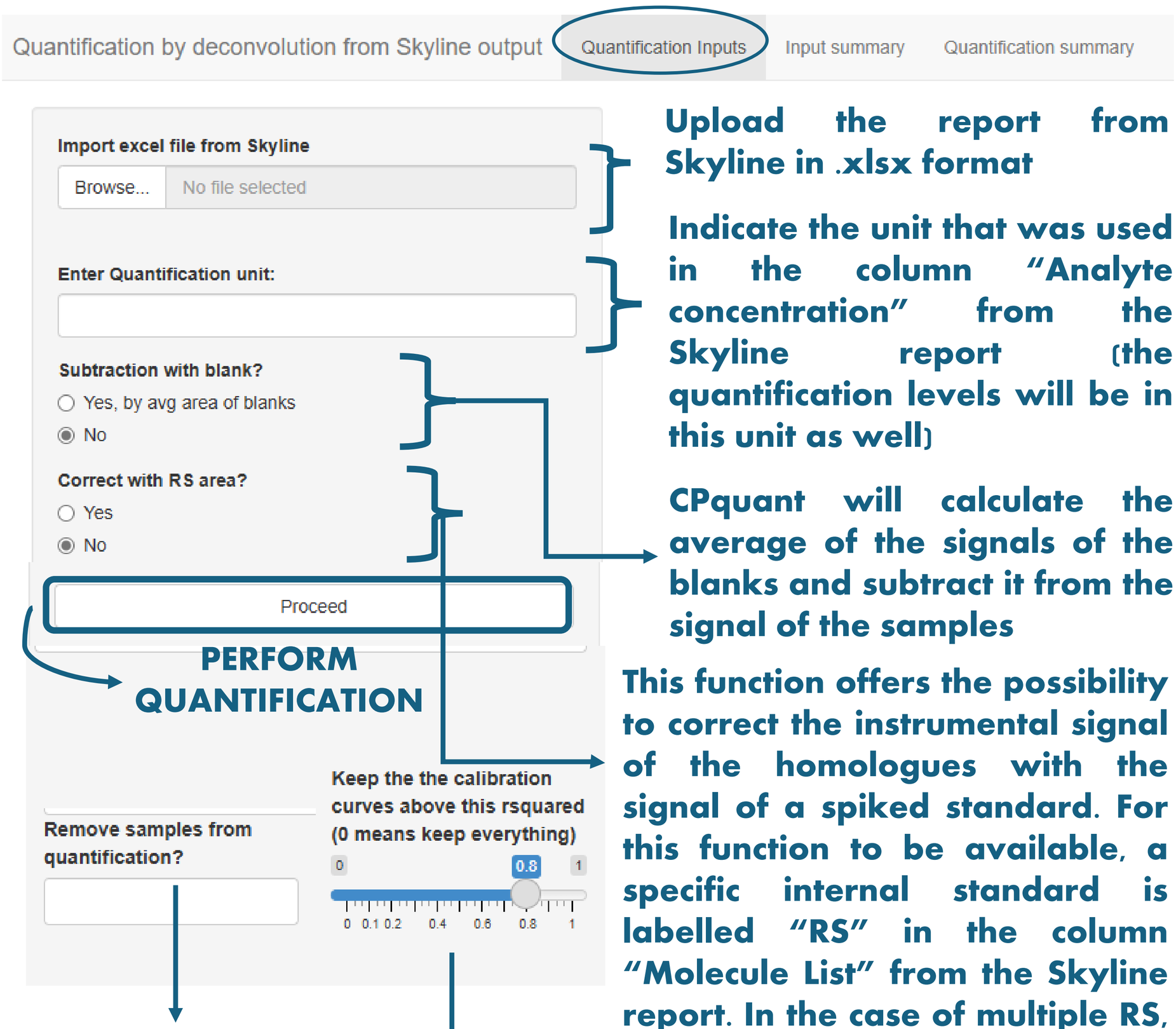
CPquant

QUANTIFY THE DATA

CPquant is a module in the CPxplorer package. It consists of an Rstudio based app whose code source can be found at https://github.com/WBS-TW/CPxplorer. A browser-based interface is available at https://wbs-tw.github.io/CPquant_Shinylive/ which can be run on the local computer without installing the R shiny app. More instructions can be found in the app (under the tab Instructions).

CPquant is a tool for performing automatized quantification of PCAs. It is compatible with the report exported from Skyline and it does not require data adaptation.

CPquant performs the homologue deconvolution (Bogdal et al., 2015; Perkons et al., 2019)[1,2]. Briefly, CPquant executes non-negative least squares regression to combine various standards and match them with the PCA pattern found in the sample. Note that CPquant performs the deconvolution on the "quan" ions from the Skyline input.



Upload the report from Skyline in .xlsx format

Indicate the unit that was used in the column "Analyte concentration" from the Skyline report (the quantification levels will be in this unit as well)

CPquant will calculate the average of the signals of the blanks and subtract it from the signal of the samples

This function offers the possibility to correct the instrumental signal of the homologues with the signal of a spiked standard. For this function to be available, a specific internal standard is labelled "RS" in the column "Molecule List" from the Skyline report. In the case of multiple RS, CPquant allows the selection of which one to use

A calibration curve is built for each standard mixture, This bar offers the option to set a minimum R^2 for the linear regression

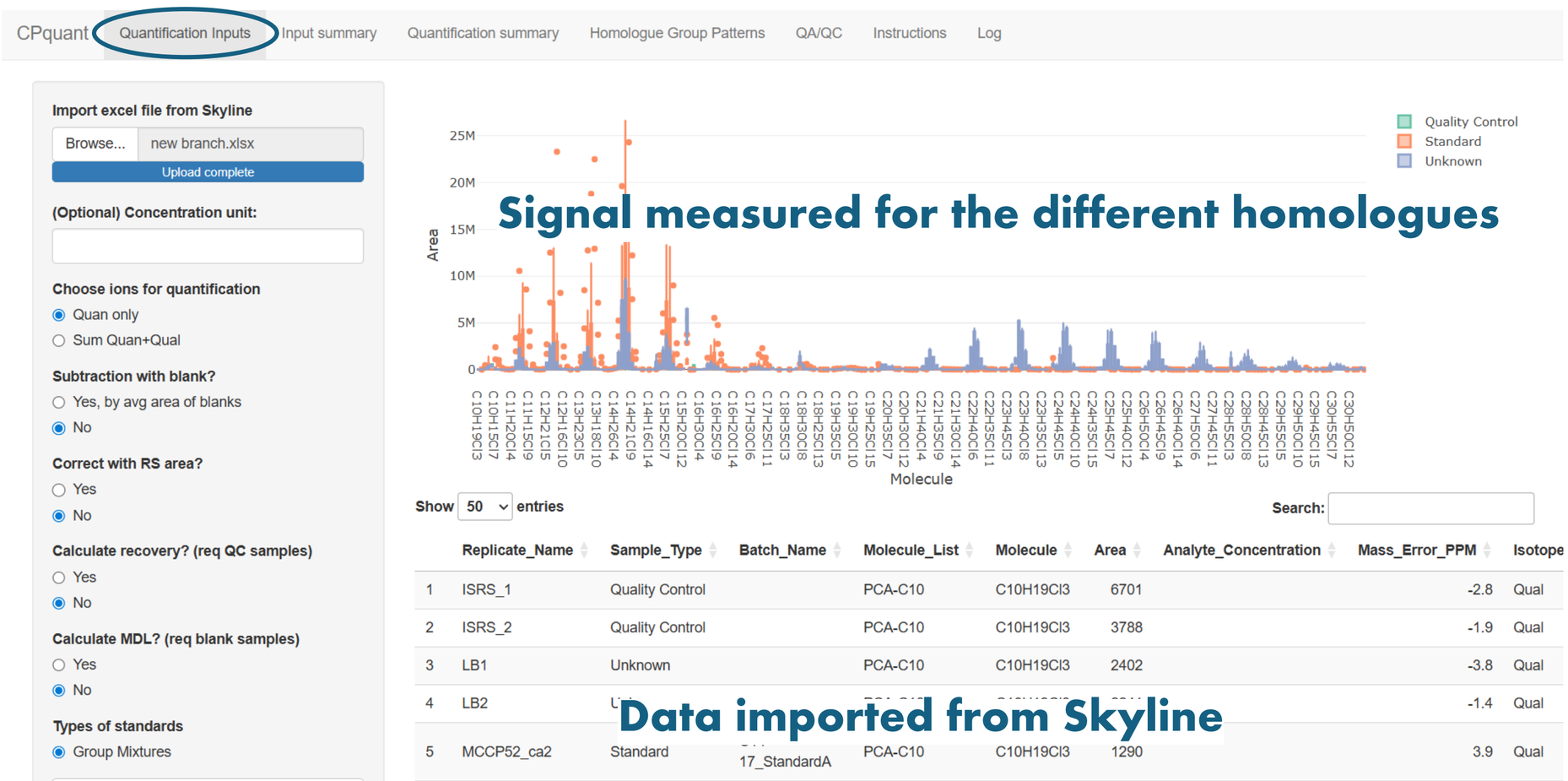
Exclude samples from being quantified

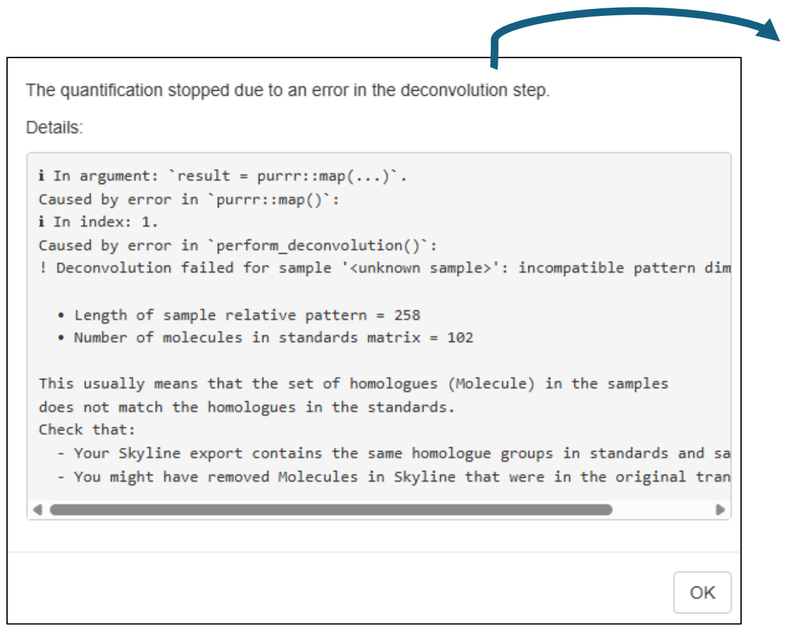
[1] Bogdal, C., Alsberg, T., Diefenbacher, P. S., Macleod, M., & Berger, U. (2015). Anal. Chem., 87(5), 2852–60. https://doi.org/10.1021/ac504444d

[2] Perkons, I., Pasecnaja, E., & Zacs, D. (2019). Food Chem., 298, 125100. https://doi.org/10.1016/j.foodchem.2019.125100

QUANTIFY THE DATA

The inputted data can be explored in the "Quantification Inputs" window. Here the data is displayed in an interactive graph and table:





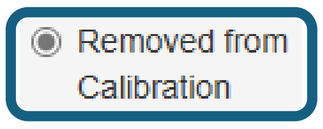
If the excel report from Skyline has a mismatch between the set of homologues (Molecule) in the samples and the homologues included in the standards, an error message will appear. For example, if the Batch_Name of the standards do not include all the homologues included as molecules, this error will appear.

The user should double check if the set of homologues matches the sta ndards included in the Skyline report, or if any molecules have been d eleted in Skyline that were present in the original ion list.

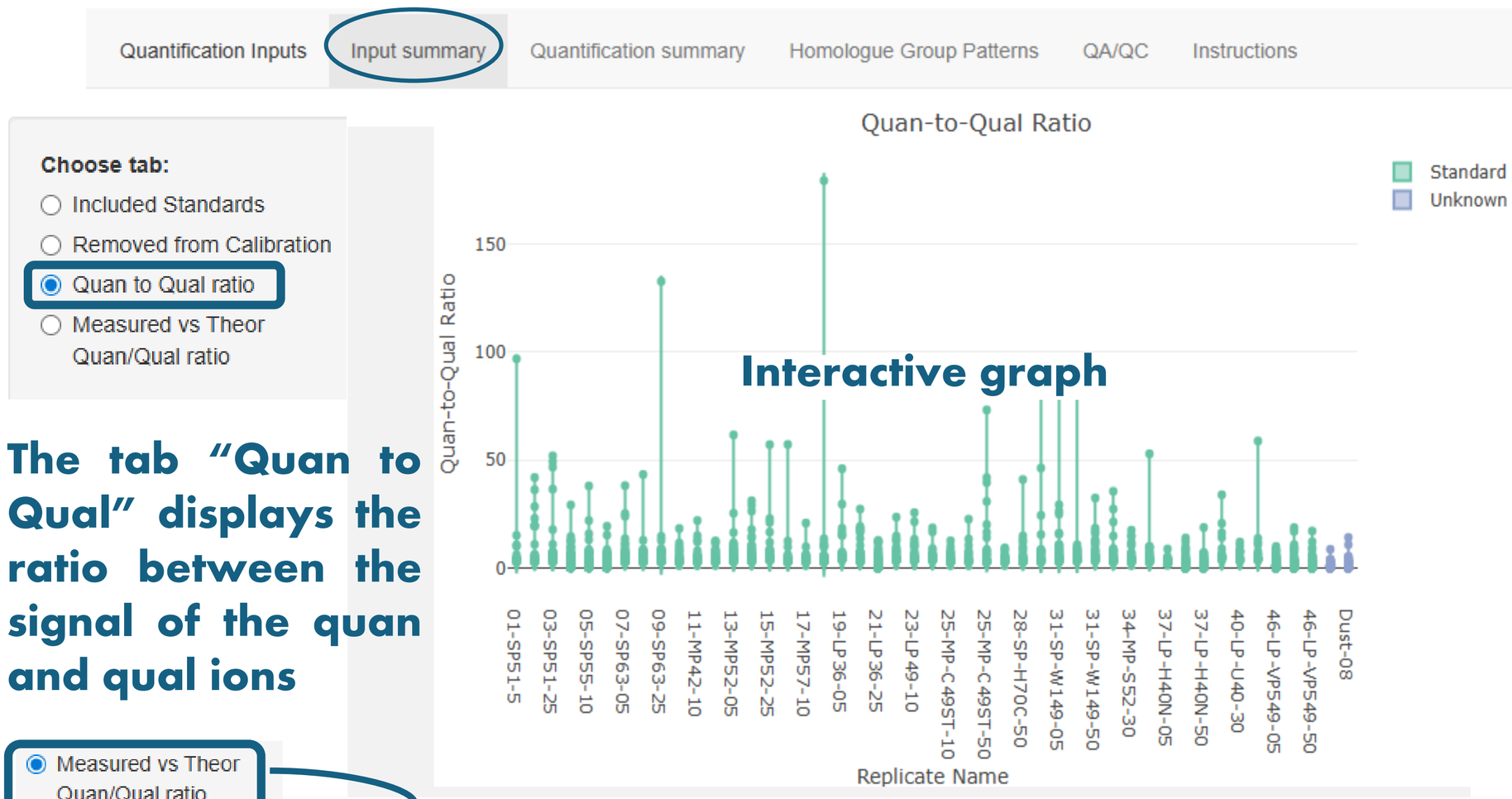
Once this is fixed and the range of homologues in the molecule list ma tches the range included with the standards, the user can proceed wit h the quantification. A confirmation message will appear:

QUANTIFY THE DATA

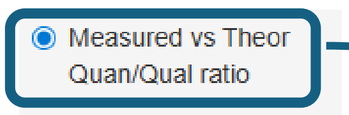
The quality of the data can be further explored in the "Input Summary" tab once the quantification has been performed. The user can consult which standard solutions have been used for building the calibration curves used in the quantification under the "Included Standards" tab.



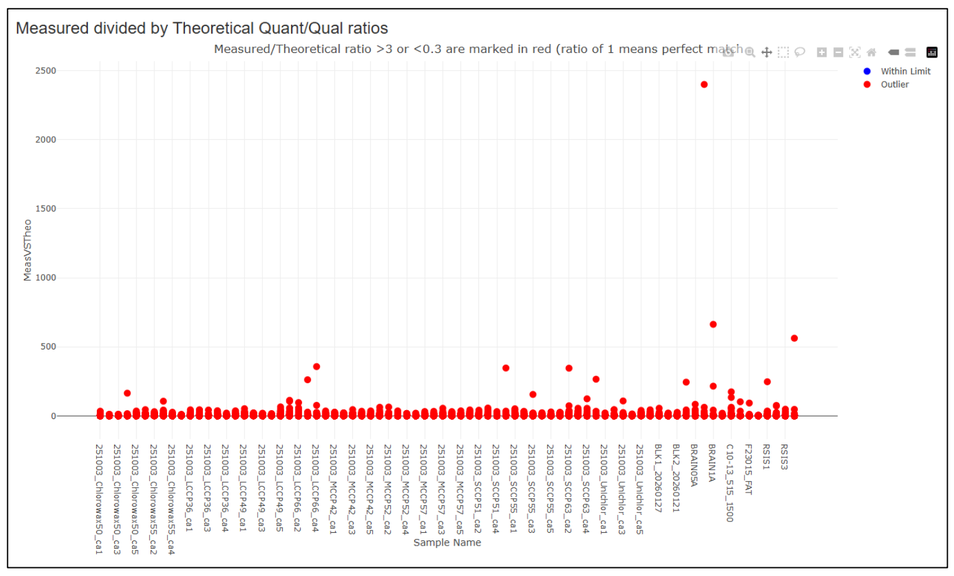
The tab "Removed from Calibration" displays homologue groups from standards that have R^2 lower than the set limit value, and are thus removed from quantification.

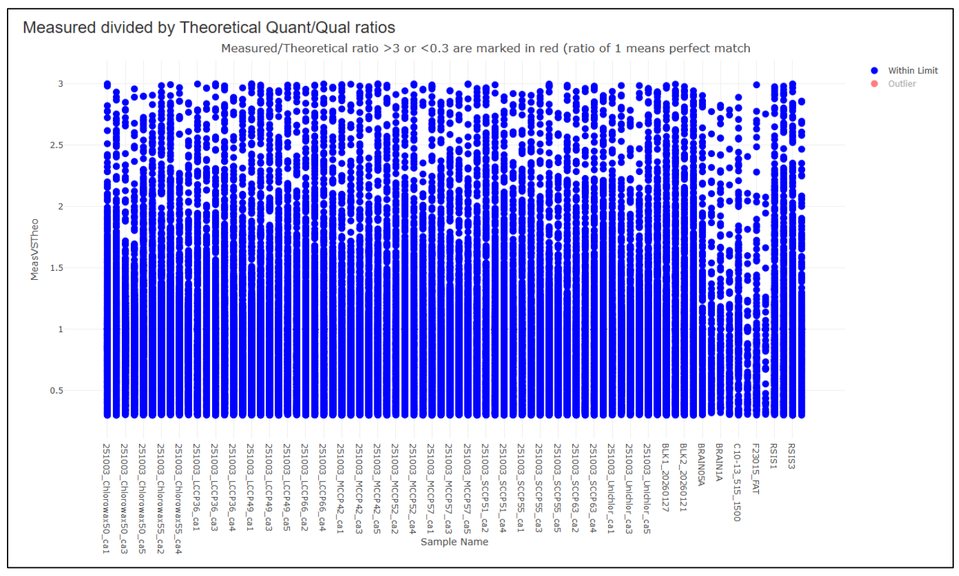


The tab "Quan to Qual" displays the ratio between the signal of the quan and qual ions



The "Measured vs Theor Quan/Qual ratio" shows a plot showing the measured Quan/Qual ratio divided by the theoretical Quan/Qual ratio. Ideally, the ratio should be around 1. Outlier ratios (<0.3 or >3) are marked in red.

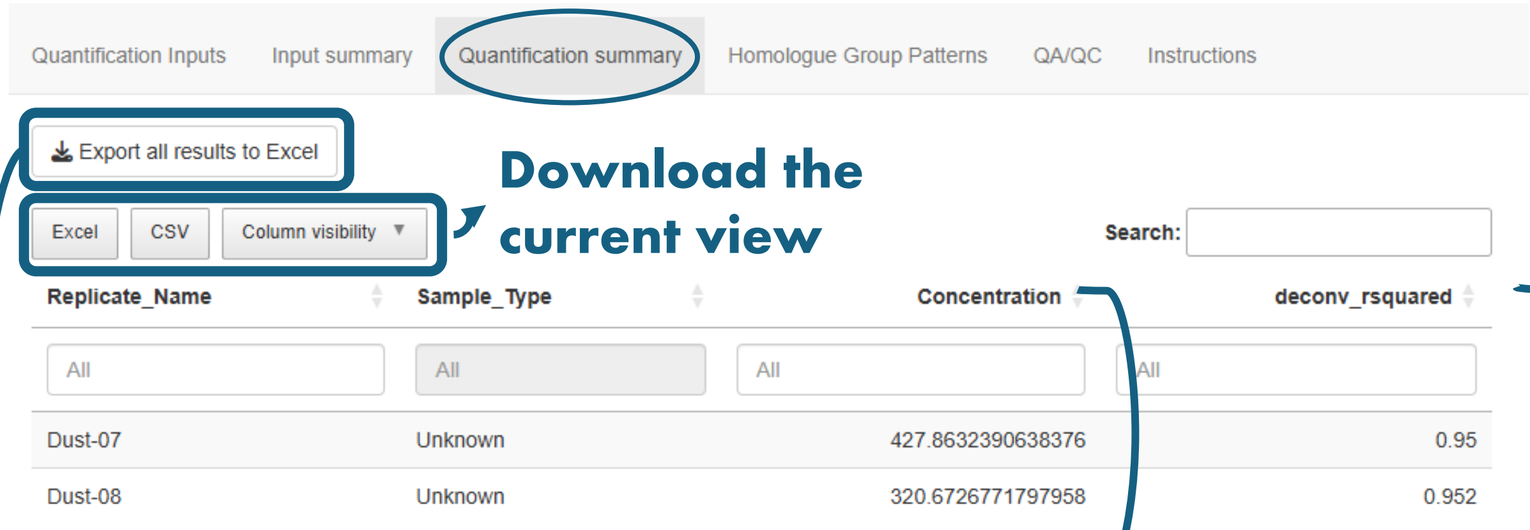




The threshold for outliers (>0.3 - <3) is arbitrary. The lower the threshold from CPions is (Isotope rel ab threshold, see page 1), more qual ions will be considered in the target list, which may lead to more outliers due to low intensity ions. If all the qual ions of one homologue group are outliers, the integration should be double-checked in Skyline to ensure the area is well integrated and it is the correct peak, however, it is advised to check the integration of any ion that shows ratios as outliers

QUANTIFY THE DATA

The quantification results can be found in the "Quantification Summary", more detailed information is displayed in the "Homologue Group Patterns" tab.



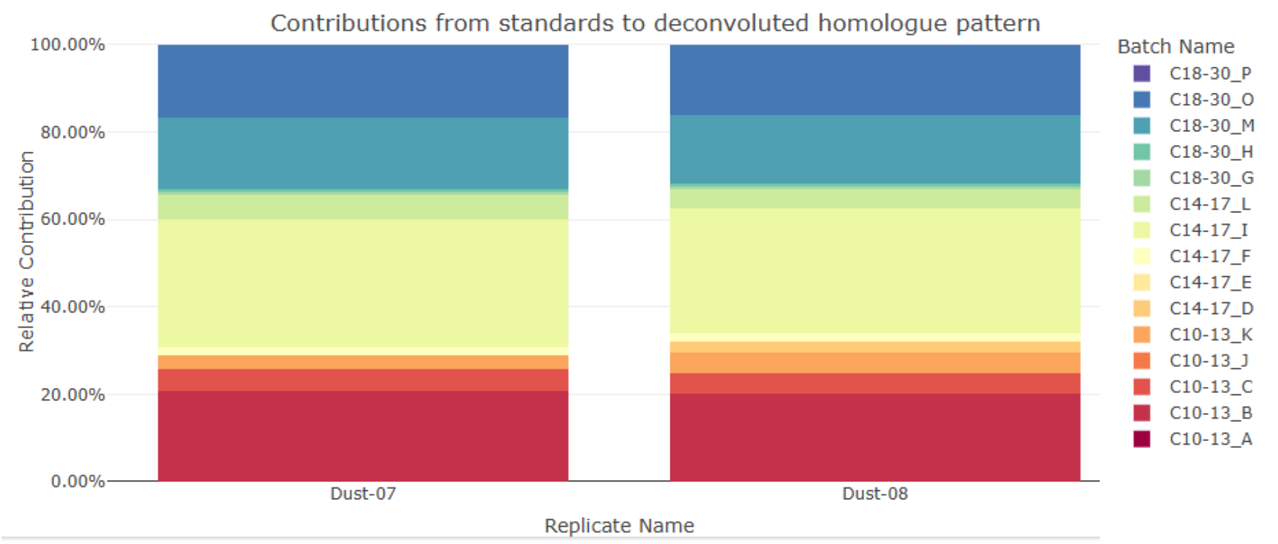
Download the current view

Download the full quantification results:
- **SumPCAs**
- **Deconvolution fit**
- **Homologue concentration**

SumPCAs

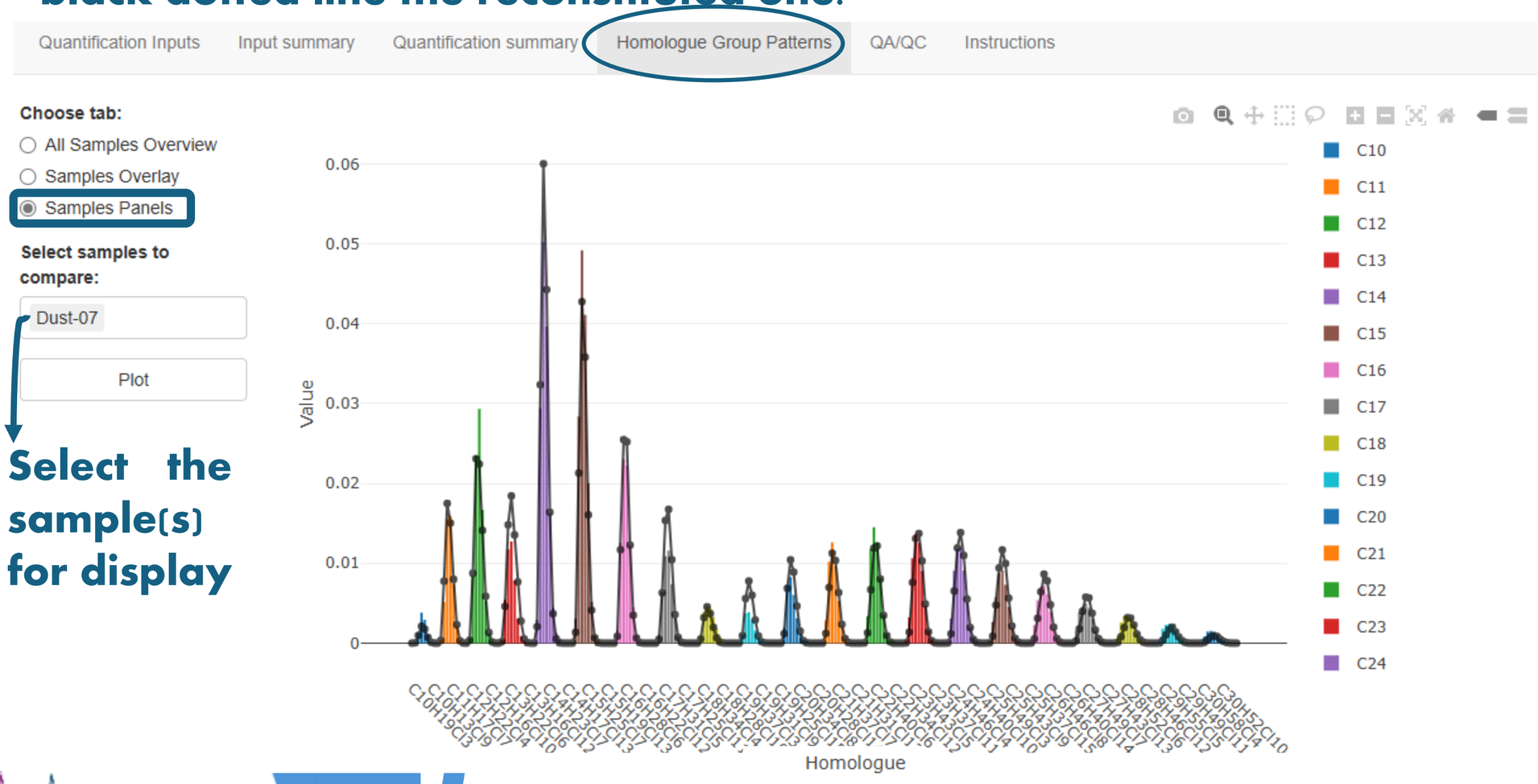
It shows how well the reconstituted pattern fits the one in the sample, where 1 would mean a perfect fit

NOTE: The concentrations correspond to the levels in the extracts



The "Standard contributions" tab displays visually the contribution of each standard for building the reconstituted pattern in the sample

A visual comparison between the reconstituted and measured patterns can be displayed at the "Homologue Group Patterns" tab where the bar plot shows the measured relative distribution and the black dotted line the reconstituted one:

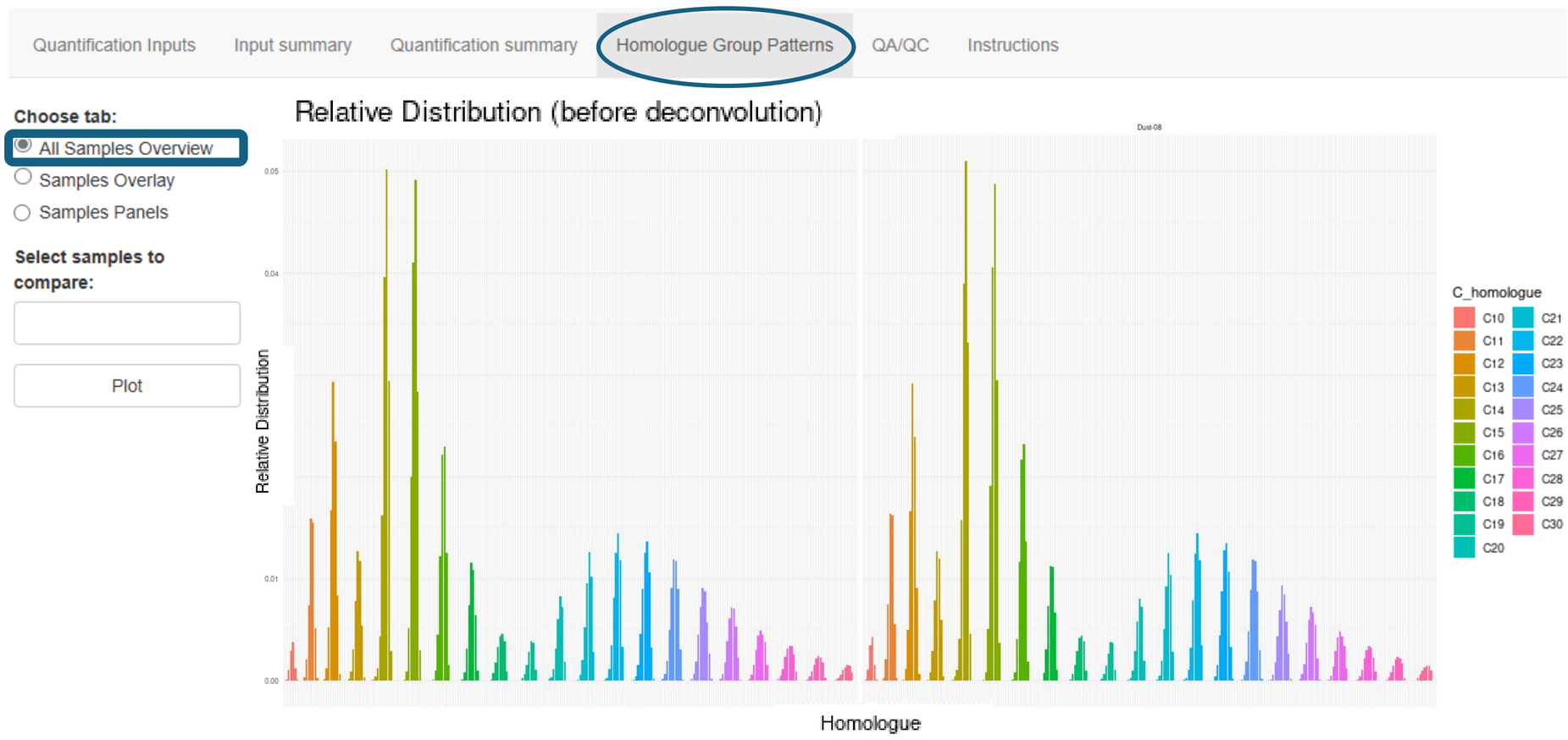


Select the sample(s) for display

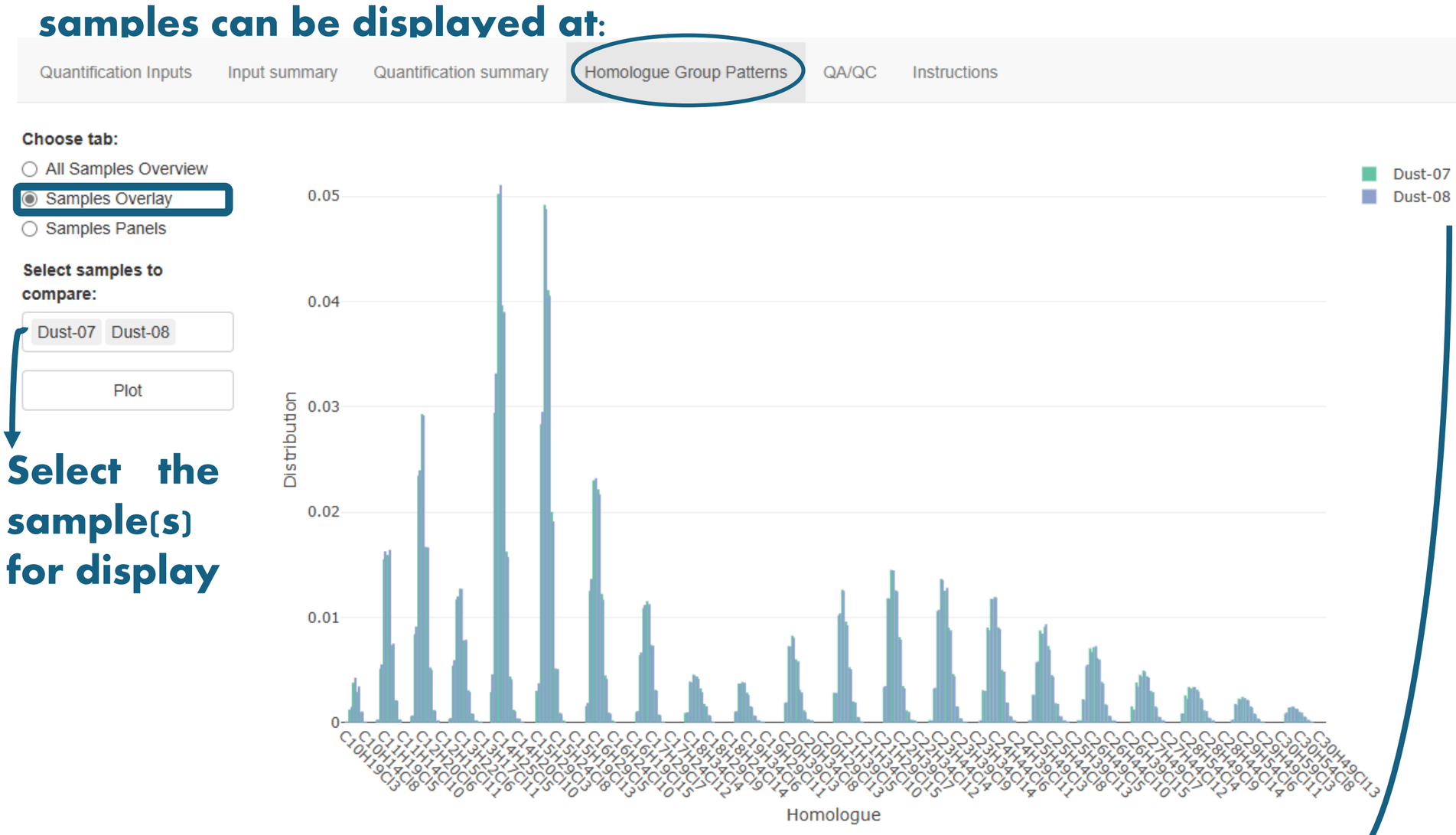
QUANTIFY THE DATA

Several displays of the homologue patterns can be found in the "Homologue Group Patterns" window.

The inputted relative distribution of the homologues in all the samples can be displayed at:



The reconstituted relative distribution of the homologues in all the samples can be displayed at:



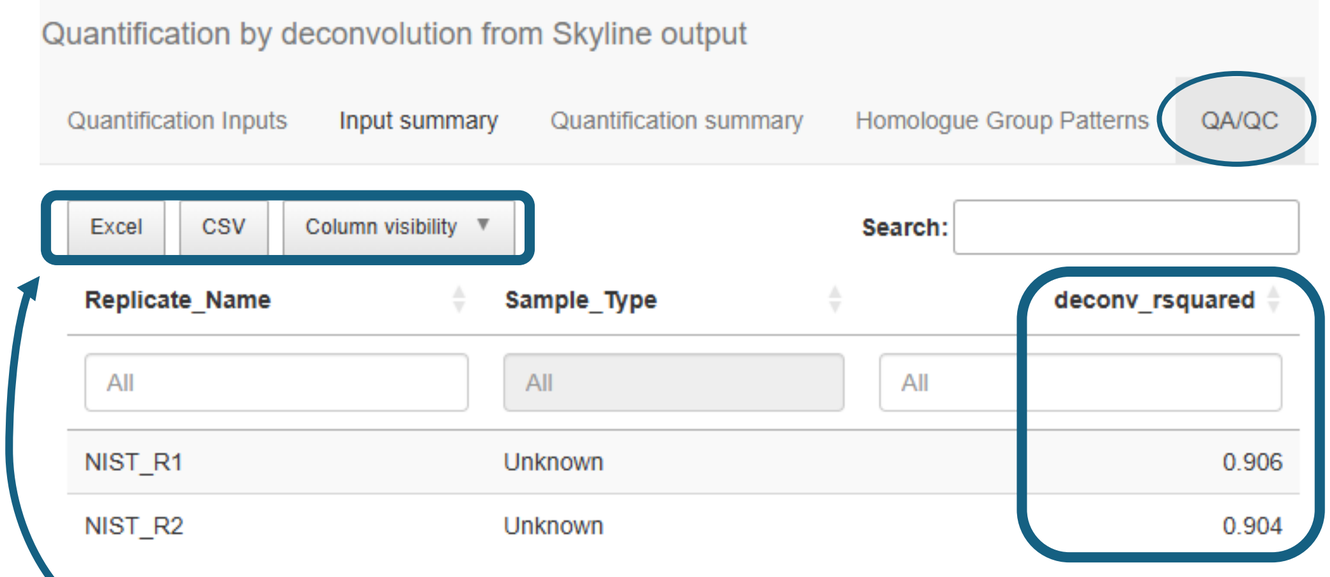
Select the sample(s) for display

The display shows several samples simultaneously and facilitates the comparison between replicates

If the user wants to plot one panel for each selected sample, it can be done on the "Samples Panels" tab. Also compares the relative distribution of homologue groups of the sample with the reconstructed pattern by the deconvolution process (as scatter lines in the Deconvoluted Distribution legend group).

QUANTIFY THE DATA

The "QA/QC" displays several parameters that serve to investigate the quality of the data.



"deconv_rsquared" shows the goodness of fit of the nnls model to reconstitute the homologue pattern of the sample

Download the report including the QA/QC results

- **<u>MDL: Method limit of detection:</u>**

CPquant can calculate the method limit of detection (MDL). It determines MDL based on the signal of the blank, therefore in order to have the MDL feature available in CPquant the report from Skyline needs to contain the files for the blank.

MDL will be calculated according to:

- **If the signal was NOT blank subtracted (this is selected in the "Quantification inputs" panel, prior to deconvolution):**

MDL = average of blank levels + 3 * standard deviation of blanks

- **If the signal was blank subtracted (this is selected in the "Quantification inputs" panel, prior to deconvolution):**

MDL= 3 * standard deviation of blank samples

- **<u>Recovery:</u>**

CPquant can calculate the recovery by using the signal of the internal standard. Therefore, the internal standard that is used for calculating the recovery needs to be named as "IS" in the "Molecule List" of the Skylilne report.

This feature is designed for a data set that contains a reference standard with the same internal concentration as the samples. This standard will be assigned as "Quality Control" for the "Sample Type" in Skyline. CPquant expresses the recovery as the ratio between the sample and the "Quality Control".

CPquant considers the signal of the "Quality Control" as a 100% recovery and calculates the ones in the sample accordingly. If the Skyline report contains more than one "Quality Control" CPquant will consider the average of their signals for the recovery calculation.

Note: If the in "Quantification inputs" panel "Correct with RS area" is selected as "YES" the signal of the internal standard used for recovery calculation will also be corrected.

CPxplorer

https://github.com/WBS-TW/CPxplorer

OVERVIEW

CPxplorer v.1.0

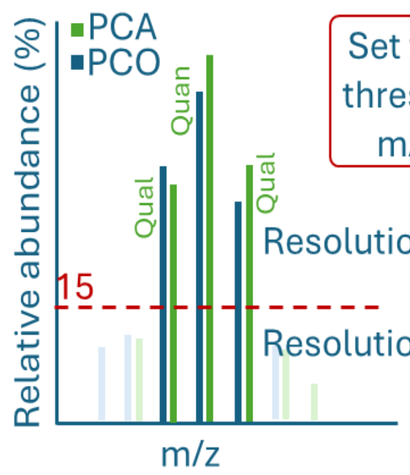
A CPions

Input parameters

Number of C	10-30
Number of Cl	5-15
Adduct	$[M+Cl]^-$
Resolution	60 000
Molecule(s)	PCA/O
Rel. abundance	15

Centroided m/z values →

Study mass spectral interferences



Set the Rel. abundance threshold to include the m/z in the target list

$$\text{Resolution} > \frac{m/z_{PCA}}{(m/z_{PCA} - m/z_{PCO})}$$

$$\text{Resolution} < \frac{m/z_{PCA}}{(m/z_{PCA} - m/z_{PCO})}$$

Create the target list →

Export the transition list

Ion	m/z	Label type
PCA + Cl-	528.9910	Qual
PCA + Cl-	530.9880	Quan
PCA + Cl-	532.9851	Qual

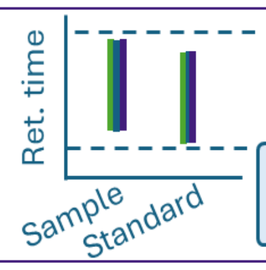
B Skyline

Import the transition list into Skyline

$C_{18}H_{31}Cl_7$: target m/z
- 528.9910 (qual)
- 530.9880 (quan)
- 532.9851 (qual)

→

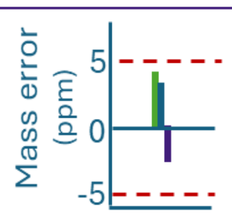
Same ret. time range



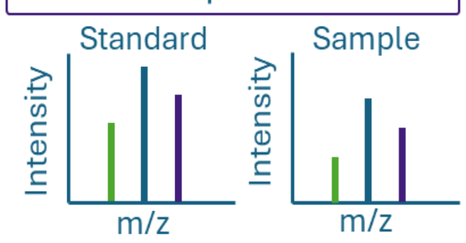
Integrate the data

Same peak shape for all isotopomers

Mass error < ±5 ppm



Same isotopic distribution



→

Export the signals report from Skyline

File	Molecule	Signal	Type	Batch	Conc (ng)
DustR1	C18Cl7	38943	Sample		
L36_1	C18Cl7	26768	Standard	C18-30_A	345
DustB	C18Cl7	04536	Blank		

C CPquant

Import the report from Skyline into CPquant

Molecule	m/z	Label type
C18Cl7	528.9910	Qual
C18Cl7	530.9880	Quan
C18Cl7	532.9851	Qual

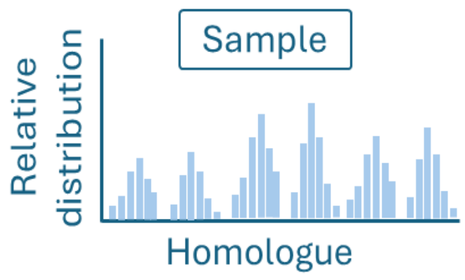
Filter the data for the signals of the quan isomers

→

Standard A

Standard B

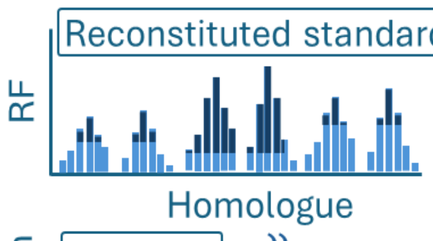
Sample



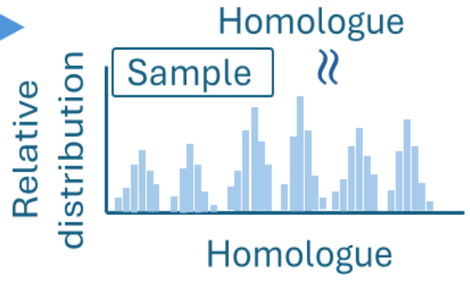
→

Perform nnls to calculate the contribution of each standard

Reconstituted standard



Sample ≈



Quantify the unknown

Sample	Molecule	Conc (ng)
DustR1	C18Cl7	250
DustR2	C18Cl7	245
DustR3	C18Cl7	255

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GOOD LUCK WITH THE DATA ANALYSIS!

Please contact our team if any questions

CPxplorer

https://github.com/WBS-TW/CPxplorer

V.1.0

A platform-independent toolkit for rapid quantification and harmonization of polychlorinated alkanes

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